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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Academic Year (2024-25 OddSemester)
HANDOUTS
Subject Code/Name: 20EC7020E -RFID and Flexible Sensors Year/Sem: IV/VII

UNIT-I
INTRODUCTION AND RFID ARCHITECTURE

Case for RFID:

- RFID technologies offer practical benefits to almost anyone who needs to keep track of physical assets.
- Manufacturers improve supply-chain planning and execution by incorporating RFID technologies.
- Retailers use RFID to control theft, increase efficiency in their supply chains, and improve demand planning.
- Pharmaceutical manufacturers use RFID systems to combat the counterfeit drug trade and reduce errors in filling prescriptions.
- Machine shops track their tools with RFID to avoid misplacing tools and to track which tools touched a piece of work.
- RFID-enabled smart cards help control perimeter access to buildings.

Some of the benefits of RFID include the following:

Alignment is not necessary:

- A scan does not require line of sight. This can save time in processing that would other- wise be spent lining up items.

High inventory speeds:

- Multiple items can be scanned at the same time. As a result, the time taken to count items drops substantially

Variety of form factors:

- RFID tags range in size from blast-proof tags the size of lunch boxes to tiny passive tags smaller than a grain of rice. These different form factors allow RFID technologies to be used in a wide variety of environments.

Item-level tracking:

- Ninety-six-bit RFID tags provide the capability to uniquely identify billions of items

Rewritability:

- Some types of tags can be written and rewritten many times. In the case of a reusable container, this can be a big advantage.

Advantages of RFID

- Cost effective solution compared to other technologies
- Does not requires direct line of sight to operate
- RFID readers can read hundreds of tags simultaneously within seconds
- RFID tags can be rewritten and easily reused
- Data from tags can be encrypted for enhanced security
- Tags can store more information than just basic product information (serial number, lot number manufacturing date, expiry, and website URL etc…)
- RFID systems can be integrated with other existing systems
- RFID technology is easily scalable and easy to implement

Eras of RFID

The progress of RFID adoption divides naturally into following eras:

- Proprietary Era
- Compliance Era

- RFID-Enabled Enterprise Era
- RFID-Enabled Industries Era
- Internet of Things Era

Proprietary Era

- For almost 60 years now (triggered by the development of transistors in 1947), businesses and governmental entities have used RFID to track items and provide access control to facilities.
- The smaller size and greater durability of transistors made it possible to attach transmitters to valuable items, and over time developments such as improved batteries, integrated circuits, and microchips reduced the cost of the transmitters (tags), allowing tracking of less valuable items.
- Some of the applications in this era included the tags used to track rail cars and the chassis tags that have been used since the 1980s to track automobiles through an assembly line. Inthe 1970s and 1980s, RFID was used for tracking dairy cattle.
- Expensive, proprietary RFID tags, which were usually recycled, were a major characteristic of this era.
- The reuse spread the cost out, such that a single use might cost only a few cents.
- Some of the systems developed during this era were technically advanced and tightly integrated into business processes, but they were characterized by both poor support for sharing information between trading partners (incompatible IDs, for instance) and costly reader and tag components.

Compliance Era

- The steep drop in semiconductor prices and widespread adoption of broadband networking at the end of the 20th century triggered an era we call the Compliance era.
- In this era, the U.S. DoD and large retailers such as Wal-Mart and Tesco began asking their suppliers to tag pallets (and sometimes individual items) with RFID tags.
- Their mandates required that the tags conform to emerging standards.
- The anticipated volume of tags that will be purchased to meet these mandates has pushed these same standards much closer to universal adoption, which has greatly reduced the cost of components.
- The new, less expensive tag technology is still prone to manufacturing defects, and, due in part to early implementation of the tag standards, often Compliance-era tags do not perform as well in practice
- Thus, while adoption of RFID is on the rise, there has actually been a slight slump in its capabilities (in the sense of how the technology is used, if not the purposes to which it is applied).
- Fortunately, new, more efficient standards and rapid improvement in both tag quality and throughput are quickly closing the gaps in the Proprietary-era systems.

RFID-Enabled Enterprise Era

- As standards stabilize and component costs fall, many organizations will begin to implement RFID tracking within their internal processes.
- This will allow them to measure the pulse of their distribution systems for materials, assets, and products and to keep real-time inventories of items, such as the location and age of perishable goods.

- During this era, declining costs will inspire a steady transition from tracking shipping units to tracking individual items.
- Other types of sensors will join RFID to monitor information such as the highest temperature to which an item has been exposed or whether gases produced by spoilage are present.
- Labor-intensive bar code inventories will, in many cases, be replaced with scan-by inventories, allowing someone simply to walk down the aisle with a handheld reader.
- Similarly, portal readers at the door will record the entry and exit of every item in a shop or warehouse.

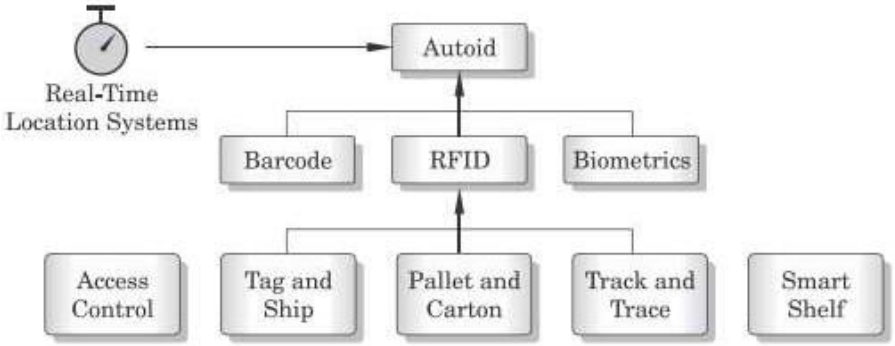
RFID-Enabled Industries Era

- In this era, RFID standards, RFID information networks, business agreements, and comprehensive security and privacy policies will solidify to the point where entire industries and supply chains can share appropriate information reliably, trusting that only authorized users can see any sensitive information.
- Safety overstock inventories will drop, along with fulfillment times and costs, due to theft and error. Simply knowing "what was where when" provides a powerful tool for applications that we have only begun to realize.
- Expect to see new products, partnerships, and even whole cottage industries develop by harnessing the advantages of this new flow of information.

Internet of Things Era

- This final era will be triggered by widespread adoption of RFID technology and the associated demand for easier management of distributed sensor networks, as well as by a reduction in the cost of smart devices and tags.
- Lower costs and greater demand for information will commercialize existing technologies already in use so that military and manufacturing applications can create self-organizing networks of cheap, expendable components with extremely low incremental maintenance and management costs.
- This technology will finally make it possible to adopt RFID technology on retail floors, in farm fields, and in homes.
- It will expand the group of businesses adopting the technology to include even the smallest entrepreneurs.
- RFID tags will cease to simply be labels applied to items and will more often be added as integral parts at the time of manufacture or as part of the packaging.
- At this stage of development, the idea that an item has a digital identity will become as basic as the idea that an item has a color or weight or size.
- In this era, physical objects will be tied to the Internet through their digital identities.

RFID applications:



Access Control:

- Access control applications are RFID systems used to selectively grant access to certain areas.
- Example: RFID tags attached to an automobile or held in a person's hand as a card, key chain, or wristband may allow access to a road, building, or secure area.

Considerations

- Anti-counterfeiting: Counterfeit tags must be recognized and attempts to use or manufacture them discouraged.
- Tailgating: Tailgating occurs when an unauthorized person or vehicle enters just behind an authorized person or vehicle before the gate or door can close.

Tag and Ship:

- Tag and ship applications are minimal RFID systems that allow a user to associate an RFID tag with an item, apply the physical tag to the item, and then verify that the tag operates properly while attached to the item.
- In some cases, these systems even use pre-encoded tags to further reduce cost.

Considerations

Cost: Because the drivers for this sort of application typically comply with a mandate, keeping cost low is the primary concern of the end user. This includes initial cost and the total cost of ownership (TCO) over time, from upgrades and repairs to monitoring and maintenance.

Pallet and Carton Tracking:

- One of the most commonly mentioned forms of RFID, pallet and carton tracking, essentially puts a "license plate" on a shipping unit made up of one or more individual items.

Considerations

- Pallet or carton integrity: This type of tracking works best with a shrink-wrapped pallet that contains only one type of item; the pallet ID is then associated with a simple item count.
- It also works well with a mixed pallet that has a more complex manifest, describing counts for more than one item.
- Pallet and carton tracking can be ineffective if there is a possibility that the pallet or carton may be broken down and reconstituted.
- In this case, the counts or manifest may become invalid.

Track and Trace:

- One of the earliest uses of RFID was to track dairy cattle.
- Now, companion animals and livestock of all types are routinely tagged with injectable glass capsules or button ear tags.
- These tags are used to identify lost pets and to sort, care for, and track the history of livestock.
- In recent years, RFID has also been increasingly used to track produce and pharmaceuticals.
- Information from livestock, produce, or pharmaceutical tracking can be critical in the event of a public health threat.

Smart Shelf:

- A smart shelf system is a set of shelves, or some other container (such as a refrigerator), that constantly keeps track of the individual items it contains.
- If an item is removed or added, the shelf immediately updates the inventory.

- By tying the identity of an item to its attributes, such as expiration date or lot number, a system using smart shelves can immediately locate all expired products and products from a certain lot.
- An example of a smart shelf system is a system that contains indicators such as horns or lights that warn users if a product has been removed from refrigeration for too long and should therefore be discarded.
- Similarly, if the user removes two drugs while are known to interact negatively, the system signals a warning.

Confluence of Technologies

Advances in semiconductor technologies

- RFID would have remained a niche technology if it was not for Moore's law and the ability of the semiconductor industry to produce chips that package processing power at a level that makes it affordable for the mass RFID market.

Intelligent devices:

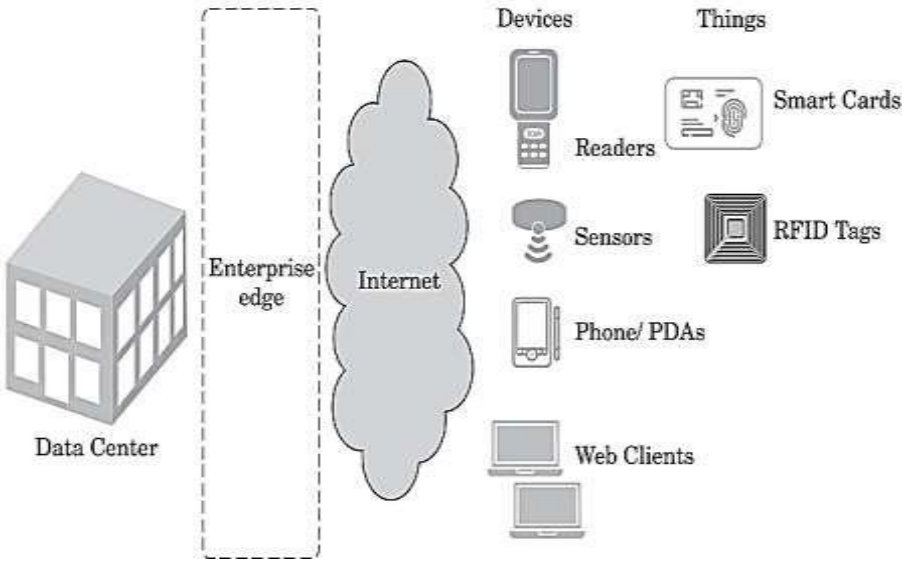
- Advances in semiconductor technologies haven't just brought down the cost of RFID chips—they are also the primary drivers behind the development of intelligent devices, including sensors such as RFID readers.
- Smarter devices and virtually ubiquitous bandwidth have opened up a host of mobility and edge-based applications.
- RFID is one implementation of the general idea of a "Network of Things" connected together to provide automation beyond the edges of corporate data centers.
- Smart homes, smart cars, and other smart objects are additional applications that require processing at the edges.
- Current implementations of smart home systems incorporate a variety of IP-enabled household devices connected to residential gateways that are in turn connected to the Internet.

Broadband wired and wireless networks and cheaper edge processing servers

- The availability of pervasive broadband data networks, coupled with affordable yet powerful servers, has led to the development of architectures that move processing to where the business processes are carried out.
- This means that it is now easier to deploy pieces of enterprise applications in edge locations such as warehouses and stores.

Edge processing capability

- Edge processing capability derives from having powerful yet low-cost personal computers and servers deployed at the edges of the enterprise network as well as a broadband connection to the data center.
- RFID systems put greater computing, data management, and bandwidth requirements on these edges.
- This is not a unique phenomenon, however, but a continuation of the overall trend.
- By "edge," we mean any location where business processes are carried out that is outside the data center or central office (for example, on production lines, in warehouses, or at retail locations).



Service-oriented architecture

- Successful adoption of RFID technologies in your enterprise will depend on how well you integrate the RFID data into your business processes.
- RFID readers can generate a lot of data.
- If it is exposed unfiltered to downstream applications, it can overwhelm them.
- To prevent applications from being flooded with RFID data and to isolate them from physical devices such as readers and antennas, you can use sophisticated middleware components such as event managers.

Key Functionalities:

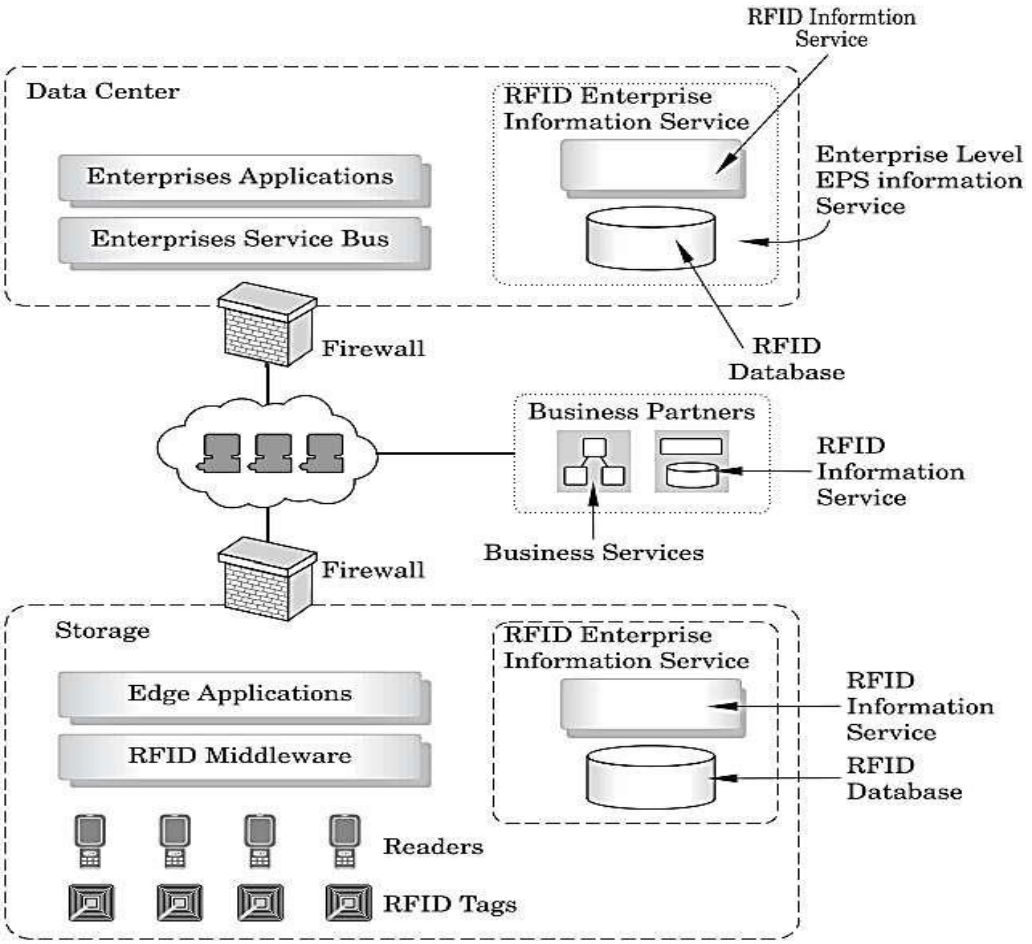
RFID system must provide the following features and capabilities:

- The ability to encode RFID tags
- The ability to attach encoded RFID tags to items
- The ability to track the movement of tagged items
- The ability to integrate RFID information into business applications
- The ability to produce information that can be shared between businesses
- The ability to develop self-organization of intelligent devices

RFID System Components:

Tags

- The term "RFID" is typically used to describe systems wherein a base station of some sort (a reader) is able to recognize another electronic device (a tag) using one of several possible wireless transmission mechanisms.
- These mechanisms may include microwave but not infrared or visible light systems.
- Since a reader is able to identify a particular tag, the system can claim to have identified the object to which that tag is attached.
- Tags may be housed in small plastic buttons, glass capsules, paper labels, or even metal boxes.
- They may be glued to a package, embedded in a person or an animal, clamped to a garment, or hidden in the head of a key.



Readers

- RFID readers, also called interrogators, are used to recognize the presence of nearby RFID tags.
- An RFID reader transmits RF energy through one or more antennas.
- An antenna in a nearby tag picks up this energy, and the tag then converts it into electrical energy via induction.
- This electrical energy is sufficient to power the semiconductor chip attached to the tag antenna, which stores the tag's identity.
- The tag then sends the identity back to the reader by raising and lowering the resistance of the antenna in a kind of Morse code.

RFID Middleware

- Choosing the right tags and readers and determining where to put the antennas is only the first step in building a working RFID system, because identifying items is only the first step in managing them.
- The capability to read millions of tags as they move through the supply chain and the need to tie tag codes to meaningful information will generate large amounts of data with complex interrelationships.
- One of the primary benefits of using RFID middleware is that it standardizes ways of dealing with the flood of information these tiny tags produce.

System Quality Considerations for RFID systems:

- A system's requirements come in two flavors:
    1. Functional
    2. Nonfunctional

- The functional requirements define what the system does, while nonfunctional requirements involve systemic qualities such as privacy and security, performance, scalability, manageability, extensibility, and maintainability.
- RFID systems demand vastly increased data-processing capabilities at the edges of corporate environments.
- The volume of raw RFID data that is captured and available for filtering, coupled with the need to automatically monitor and manage RFID devices (such as antennas and readers), requires a level of sophistication that was generally absent from the bar code–type systems that RFID systems replace.
- Some of the most important systemic qualities for RFID systems are
    - Privacy and Security
    - Performance
    - Scalability
    - Manageability
    - Extensibility and Maintainability

Privacy and Security:

- Privacy and security considerations are a core part of the design and architecture of any RFID solution.
- Ironically, even though RFID solutions are sometimes offered as enhancements to security, the technology itself presents certain vulnerabilities.
- As in any enterprise system, security considerations for an RFID system ensuring the authenticity of the information stored on the tags themselves, securing the transmission of information between tags and readers, and ensuring overall application and infrastructure security permeate the various layers of its architecture.

Performance:

- Performance is measured in terms of time taken to perform a unit of activity.
- Depending on the layer of the RFID system at which you're working, the performance considerations will vary.
- For instance, for the physical layer, the time required to recognize a tag is an importantperformance consideration.
- As with the performance metrics for other systems, this measure would be taken at peak throughput and averaged over a period appropriate to system usage.

Scalability:

- Scalability has to do with how small a system may start and how quickly it must be ready to grow.
- For RFID systems, scalability involves much more than "CPU headroom."
- Once readers are in place, changes to the RF environment can usually be spotted quickly due to changes in performance, but during an install, the environment offers enough variables without adding unexpected signal absorption and reflection.

Manageability:

- RFID information servers are typically housed in a data center or server room.
- Operations staff can monitor and manage the servers using procedures developed for any other application server.
- Event managers and readers are found in loading bays and warehouses, in truck trailers and train yards, and even in corrugated tin shacks in the middle of fields.

- A reader may be just as complex as a server, requiring software or firmware upgrades, monitoring, and management.
- An event manager is likely to be located on server hardware but deployed close to the readers. These are edge systems and must be managed as such.

Extensibility and Maintainability:

- For most organizations, RFID readers are the first step into a new space at the edge.
- Some industries especially within the areas of manufacturing, agriculture, and defense have worked with distributed sensors for some time, but a company new to RFID may falsely assume that once it gets the RFID readers talking to the management applications, it's done.
- Mandates and contracts that today require RFID compliance could easily requiretemperature, humidity, shock, or magnetic field exposure tracking tomorrow.

System Management

- The level and scale of automation for some RFID systems poses significant challenges for system administrators and managers.
- If a manual bar code scanner is down, a human operator will immediately pick up on it and call for service.
- The automated processing capabilities that RFID technologies offer make automated detection of system faults and stoppages indispensable.
- Similarly, we need to provide novel solutions for provisioning, monitoring, and managing our silently ticking antennas, readers, and other RFID infrastructure components.
- Think ease of configuring, provisioning, managing, and monitoring when deciding on infrastructure components such as readers, sensors, event managers, servers, storage, and networks. Build redundancy into the architecture.

Start with Architecture

- Much of the time, enterprise architecture grows out of its selection of products.
- With minor exceptions, over time, a proprietary architecture that is locked into a particular product and vendor develops.
- Begin with well-defined interfaces between different layers, such as event managers, applications, and integration servers, and then choose products that fit your architecture.
- Later, this will pay in greater flexibility and independence.

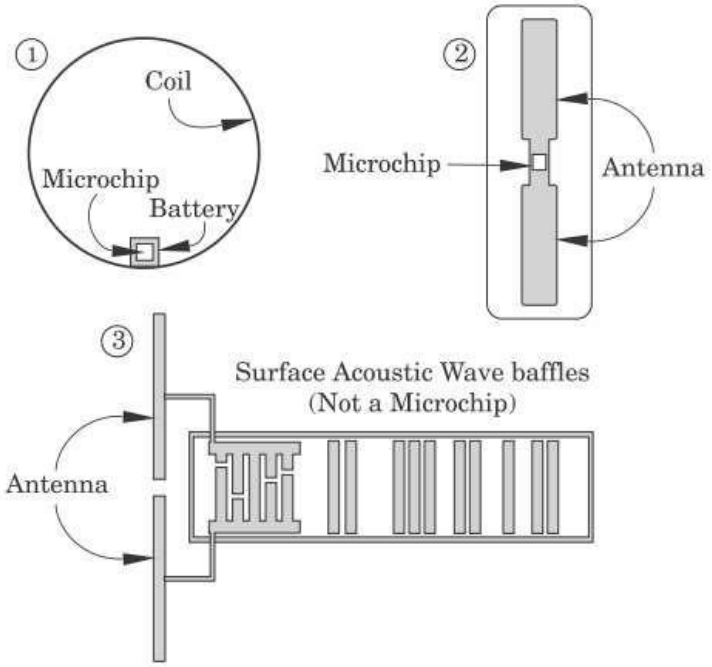
Use RFID Middleware

- It is necessary to use RFID middleware for all but trivial implementations.
- RFID middleware helps decouple your enterprise applications from the physical architecture of your RFID systems; those applications don't need to know about the reader vendors, their APIs, or the physical configuration of readers and antennas.
- In addition to this, RFID middleware provides the important function of filtering the highly fragmented data that comes from the readers.
- By cutting down the volume of data that passes over wide area networks, event filtering also allows you to define application- level events and pass more meaningful information to your enterprise applications.

UNIT-II
TAGS AND PROTOCALS

Basic Tag Capabilities:

- The purpose of an RFID tag is to physically attach data about an object (item) to that item.
- Each tag has some internal mechanism for storing data and a way of communicating that data.
- Not every sort of RFID tag has a microchip or a built-in power source, but every RFID tag has a coil or antenna of some sort.
- While it is important to realize what all tags have in common, classifying tags helps in understanding how they work.
- Many basic operations can be performed with an RFID tag, but only two of them are universal.



- The operations are
  - ➢ Attaching the tag
  - ➢ Reading the tag: Kill/disable, Write once, Write many, Anti-collision, Security and encryption, Standards compliance

Physical Characteristics

❖ RFID tags must physically attach data to items of different shapes and sizes in different environments.

❖ Some of the physical characteristics of various tags include:

➢ PVC or plastic buttons and disks→ central hole for fasteners (durable and reusable)

➢ RFID tags shaped like credit cards, which are called “contactless smart cards.”

- Tags made into the layers of paper called →"smart labels." like bar code labels.
- Small tags embedded in common objects such as clothing, watches, and bracelets. These small tags may also come in the form of keys and key chains.
- Tags in glass capsules, which can survive even in corrosive environments or in liquids.

Power Source

- Tag can be categorized by their source of power. It also determines the factors for the cost and longevity of a tag.
- *Passive* tags obtain all of their energy by some method of transmission from the reader.
- *Active* tags use an on-board battery to power communications, a processor, memory, and possibly sensors.
- Traditionally, tags that use battery power for some functions but still allow the reader to power communications have been termed "active" as well, but for clarity, we will use the more recent terminology for them: *semi-passive*.
- One additional type of tag → not only capable of supplying power for itself but is also able to initiate communications with other tags of its own kind without the aid of a reader. These tags are called *two-way* tags.

Protocol Terms and Concepts

- A protocol is defined as a set of formal rules describing how to transmit data, especially across a network.
- Low level protocols define the electrical and physical standards to be observed, bit- and byte ordering, and the transmission and error detection and correction of the bit stream.
- High level protocols deal with the data formatting, including the syntax of messages, the terminal to computer dialogue, character sets, sequencing of messages, etc.
- Technical jargon develops around any new technology, and RFID is no exception.

i. Singulation

- This term describes a procedure for reducing a group of things to a stream of things that can be handled one at a time.
- For example, a subway turnstile is a device for singulating a group of people into a stream of individuals so that the system may count them or ask them for access tokens.
- This same singulation is necessary when communicating with RFID tags, because if there is no mechanism to enable the tags to reply separately, many tags will respond to a reader at once and may disrupt communications.
- Singulation also implies that the reader learns the individual IDs of each tag, thus enabling inventories.
- Inventories of groups of tags are just singulation that is repeated until no unknown tags respond.

ii. Anti-collision

- This term describes the set of procedures that prevent tags from interrupting each other and talking out of turn.
- Whereas singulation is about identifying individual tags, anti-collision is about both regulating the timing of responses and finding ways of randomizing those responses so that a reader can understand each tag amidst the plethora of responses.

iii. Identity

- An identity is a name, number, or address that uniquely refers to a thing or place. "Malaclypse the Elder" is an identity referring to a particular person.
- "221b Baker Street London NW1 6XE, Great Britain" is an identity referring to a particular place, just as "urn:epc:id:sgtin:00012345.054322.4208" is an identity referring to a particular widget.

Standards:

- Some RFID applications need to interoperate only with the procedures and systems of a single company.
- Others must share information with a global consortium of partners.
- No matter where in this spectrum an individual application falls, standards will affect the choice of tags for that application—if only because large manufacturing runs of tags fitting a particular standard may make those tags less expensive than equivalent proprietary tags, even though the latter may be more simply designed.
- Choosing a tag is, to a large extent, choosing the type of RFID system you intend to build, and tag standards often involve much more than the physical characteristics and air interface of a tag.

EPC global Tag Types

- EPC global, Inc., a collaboration between GS1 and industry partners, defines a combined method of classifying tags that specifies frequencies, coupling methods, types of keying and modulation, information storage capacity, and modes of interoperability.
- EPC tags are intended to carry EPC numbers, which are assigned by the specific management entities who own the object classes involved.

Class	Discription
Class 0	Passive, read-only.
Class 0+	Passive, write-once but using Class 0 protocols.
Class I	Passive, write-once.
Class II	Passive, write-once with extras such as encryption.
Class III	Rewritable, semi-passive (battery-powered chip, reader-powered communications), integrated sensors.
Class IV	Rewritable, active, "two-way" tags that can talk to other tags, powering their own communications.
Class V	Can power and read Class I, II and III tags and read Class IV and V tags, as well as acting as Class IV tags themselves.

IS0/IEC 18000 Tags:

- In the past, GS1 has backed both the EPC standards and the GTAG initiative, which includes the 18000-6 standard for UHF.

- This standard originally conflicted with the EPC UHF specification and the new EPC UHF Generation 2 specification, but tag manufacturers were already selling EPC Class 0, Class 0+, and Class I tags.
- Some have expressed concern that the 18000 air interfaces and command protocols are more complex than those defined by the EPC standards, which could lead to both more expensive tags and stall adoption of the standard.

How Tag Stores Data:

- The high-level tag communications protocols know about the ID types that can be stored on a tag and so know, at a general level, how data is stored on the tag.
- However, since a reader only communicates with a tag via the air protocol, the actual physical layout of memory on a tag is left to the tag manufacturer.

- The CRC is a checksum, the EPC is the ID on the tag, and the password is the "kill code" to disable the tag.
- Note that for a tag to be EPC-compliant under Class 0 and Class I Generation 1 standards, it must never transmit the password under any circumstances not even in response to proprietary diagnostic commands used only by the manufacturer.
- This password is the same password described previously as a means of disabling (destroying) tags.

CCITT-CRC:

- A Cyclic Redundancy Check (CRC) is a way of verifying that a block of data has not become corrupted.
- The sender of the data block calculates a value by treating the whole block as one large number and dividing it by a number called the CRC polynomial.
- The remainder of this operation is the CRC.
- The sender sends this CRC along with the data, and the recipient uses the same method to calculate a CRC over the data block for comparison.
- If the CRC from the sender does not match the CRC calculated by the recipient, the recipient may request that the data be resent.

Singulation and Anti-Collision Procedures:

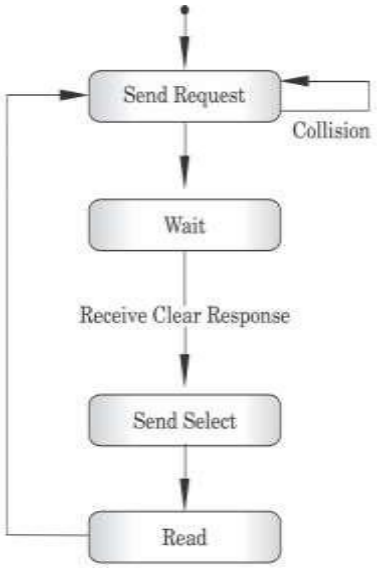
There are many different ways for readers and tags to communicate, but the different methods can all be broadly categorized as Tag Talks First (TTF) or Reader Talks First (RTF).

- Logically, it would seem simplest for a tag arriving on the scene to announce its presence to all concerned.
- In practice, however, this is difficult unless tags can negotiate among themselves which tag will speak first.
- Some high-end active tags use TTF communications protocols, but the new crop of inexpensive smart labels and other passive tags exclusively use RTF protocols.

Slotted Aloha:

- Slotted Aloha is derived from a procedure known simply as "Aloha," which was originally developed in the 1970s by Norman Abramson of Aloha Networks in Hawaii for packet radio communication.

- Aloha was the inspiration for the Ethernet protocol, and a variation of this procedure is still used for satellite communication as well as for ISO 18000-6 Type B and EPC Gen2 RFID tags.
- Aloha itself is as simple as an anti-collision procedure may reasonably be and doesn't really include the concept of singulation in the usual sense.
- With this procedure, tags begin broadcasting their IDs as soon as the reader's field energizes them.
- Each tag sends its entire ID and then waits for a pseudorandom period of time before broadcasting again.
- The reader simply receives the IDs, depending on chance to ensure that each tag will eventually broadcast during a period when all other tags are quiet.
- The reader doesn't respond to the tags in any way.
- The advantages of this procedure are speed and simplicity.
- Tag logic is minimal, and with such a low protocol overhead, this is the top performer for read rate where only a few tags are present.



Tag Features for Security and Privacy

- There are justifiable privacy and security concerns with any sort of identification technology, and RFID, as much as biometrics, has raised popular concern because of its possible impact on personal privacy.
- A common concern is that an unauthorized person will be able to obtain information from, or possibly even change information stored on, an RFID tag.
- There is some legitimacy to these concerns.
- A tag that does not use a secure protocol might be read by anyone, just as a bar code or printed label might be read by anyone.
- A writable tag might also be maliciously altered if it does not implement a secure protocol for communication with a reader.
- This alteration might be less obvious than someone placing a false bar code or printed tag on a piece of merchandise.
- Unlike printed tags and bar codes, which will always remain at risk, future secure protocols will make RFID tags very difficult to forge or even read without authorization.
- The more secure the tag, however, the more expensive it is, and an unencrypted, read-only tag is no more risky in the right application than a bar code is now.
- For these reasons, many supply chain applications currently use less-secure RFID implementations.
- As overall RFID prices drop due both to wider adoption and standardization, the cost of more secure will also drop, and these in turn will become more widely adopted.

Destroying and Disabling Tags:

- A key concern of privacy advocates is the longevity of an identification mechanism such as RFID.
- Few privacy concerns arise from tracking pallets in a warehouse, or even from tracking individual items from the warehouse to the shelf.
- The point when an end user or customer first touches an item is the point where most privacy concerns arise.
- Within the store, expectations of privacy are tempered somewhat by the current precedent of surveillance from cameras and electronic anti-theft devices.
- But when RFID tags may still be attached and even active when a product leaves the store, there is a real possibility that tracking may continue outside the store, and this raises many concerns.

Encoding Techniques of Tag:

Encoding determines the way the tag and reader will interpret changes in the analog carrier to represent digital data.

- Thus, an encoding is an agreement between sender and receiver regarding what the keyed changes mean.
- Morse Code, one example of an encoding, uses long tones to represent "dashes" and short tones to represent "dots."
- If we substituted "0" for "dash" and "1" for "dot," Morse encoding would work for sending information over a serial bus.
- It might seem odd to think of RFID as devices communicating over a serial bus, but even though the components are wireless rather than wired, most tags and readers use serial communication because of the limited frequency range available.
- RFID even uses some of the same encodings that many of us first encountered in Universal Asynchronous Receiver/ Transmitter (UART) chips.

Various encoding schemes commonly used in RFID include:

i. Biphase Manchester encoding

- This encoding uses a negative transition in the middle of the bit cell to mean a "1" value and a positive transition in the middle of the bit cell to mean a "0" value.
- Transitions that happen at the bit cell boundary do not encode a value and are used to "reset" the encoding to send the same value again in the middle of the cell.
- Encoding is independent of the type of keying or modulation, so this could be an ASK, FSK, or PSK transition.
- Because transitions are synchronized with the clock, this method is said to have the clock in the data stream.
- A recipient can derive the clock signal from this encoding and so regain synchronization, which makes biphase Manchester encoding well suited for transferring large messages.
- ISO 18000-6 type B tags use biphase Manchester encoding to talk to a reader.

ii. Pulse interval encoding

- This is similar to Morse Code, where values are encoded by the length of the pause, or interval between, short pulses.
- A pause of a specific length represents a "1," while a pause twice as long represents a "0."

- This has the advantages of requiring less power and being resistant to noise.
- It may be a problem for longer transmissions, though, because the data stream does not contain a clock.
- ISO 18000-6 type A tags use pulse interval encoding to send messages to a reader.

iii. Biphase space encoding

- In biphase space encoding a type of encoding that is often used for reader-to-tag communication transitions happen at each clock tick (actually, at the edges of the bit cell, but it amounts to the same thing).
- If a transition occurs in between these "ticks" of the clock, it represents a "0"; no extra transition represents a "1."
- This encoding method contains the clock in the data stream and so is well suited for long messages. ISO 18000- 6 readers use biphase space encoding.

iv. Pulsed RZ encoding

- This is a term we've coined to describe a type of encoding in which four short pulses in the first half of a bit cell represent a "1," while four short pulses in the last half of the bit cell represent a "0."
- The "RZ" means "return to zero," which is a term used to describe encodings in which the signal returns to a point that is neither high nor low (i.e., at zero) for some part of the bit cell.
- EPC HF readers use pulsed RZ encoding with four pulses for each symbol.

v. EPC Miller encoding

- Miller encoding encodes a number by having a transition in either direction at the half bit point.
- EPC UHF Gen2 uses a version of this encoding that can have several timing pulses in a bit, wherein a "transition" is actually represented by the lack of one of the timing pulses.
- The number of transitions to use as a timing pulse in a bit is called the "M" value.
- For example, for M=2 we would have 2 pulses per bit cell to mark the timing of the clock.
- If one of these pulses does not occur, this represents a "1"; otherwise, the bit cell represents a "0."

vi. "1 of 256" and "1 of 4"

- These two forms of encoding are used in tags that conform to ISO 15693.
- For "1 of 256," values from 0 to 255 may be encoded in the data stream by the timing of a pulse.
- There are 512 timing slots per frame during which a pulse may occur, and the value is encoded by multiplying it by 2 and then adding 1.
- This gives the slot location—for example, a value of 12 would be encoded by a pulse at slot 25.
- Only odd slots are used for data, as even-numbered time slots are used for End of Frame (EOF) signals.
- Counting the odd slots leaves 256—thus the name of this encoding and its capability to send an entire byte in each frame.
- "1 of 4" encoding works in a similar way but uses only eight total time slots, which allows for 2 bits to be sent at a time.

vii. FSK subcarrier encoding

- This is another term we've coined for want of a better one.
- This sort of encoding is the exception to our earlier claim that RFID communicates over a serial bus and that encoding and keying are different.
- In this type of encoding, two subcarriers represent "1" and "0," respectively, and the sender uses FSK to create pulses on these subcarriers in time with the data stream.
- EPC UHF tags use FSK subcarrier encoding to talk to a reader.

RFID tag's Coupling Mechanism:

- A tag's coupling mechanism determines the way a circuit on the tag and a circuit on the reader influence each other to send and receive information or power.
- The type of coupling a tag uses directly affects the read range between the tag and reader.
- We can group the different read ranges loosely into those systems where the read range is close (within 1 cm), remote (1 cm to 1 m), or long-range (more than 1m).
- A synonym for remote coupling is "vicinity coupling."

Types

1. Backscatter coupling
2. Inductive coupling
3. Magnetic coupling
4. Capacitive coupling

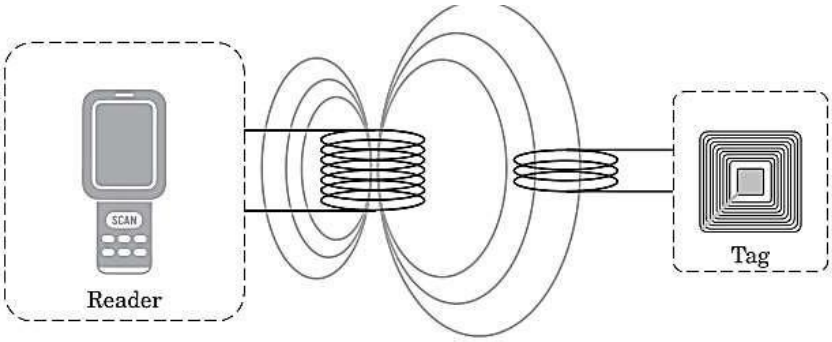
- Capacitive and magnetic coupling are examples of close coupling, inductive coupling is a type of remote coupling, and backscatter coupling may be remote to long-range.
- Along with range, the choice of coupling mechanism strongly affects which frequency the tag should use.
- Inductive coupling typically works best in either the LF or HF range.
- Backscatter coupling works better with higher frequencies.
- Magnetic coupling is most efficient at 1 – 10 MHz, and capacitive coupling usually runs at about 10 MHz to provide a fast enough clock for communications and processing.

Backscatter coupling:

- Backscatter coupling provides an elegant solution to the puzzle of how to make an RFID tag without a battery.
- The name itself, "backscatter," describes the way the RF waves transmitted by the reader are scattered back by the tag.
- The waves are reflected back to the source to send a signal.

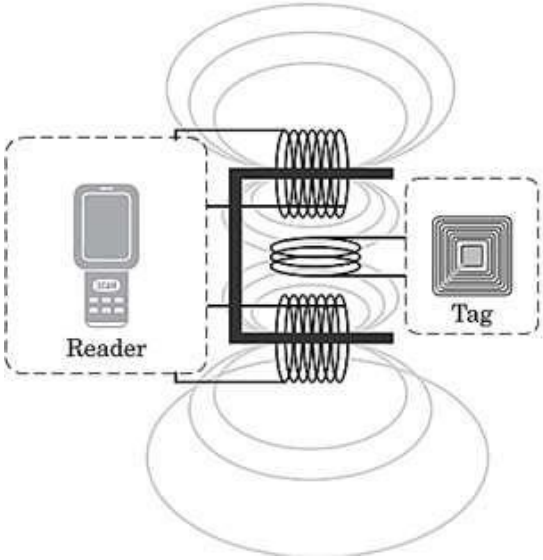
Inductive coupling

- Inductive coupling is a common type of remote coupling.
- Some of the most common types of tags in recent years have been various kinds of inductively coupled tags, including tags that follow the ISO 15693 standard for vicinity- coupled smart cards.
- A reader powers inductively coupled tags by using a coil antenna to generate a magnetic field.
- The field drives current through a coil on the tag by induction in much the same way that a transformer transfers energy between two coils.
- For this reason, this type of coupling is sometimes called transformer coupling.



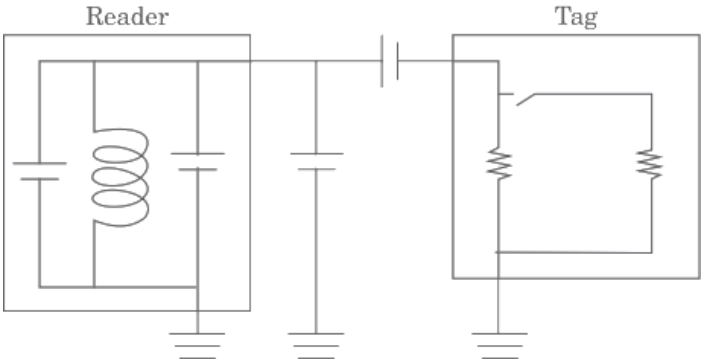
Magnetic coupling:

- Magnetic coupling is a close coupling that is similar to inductive coupling in that the readerand tag form a pair of transformer coils.
- The major difference is that the reader coil in magnetic coupling is a round or u-shaped ferrite core with windings.



Capacitive coupling:

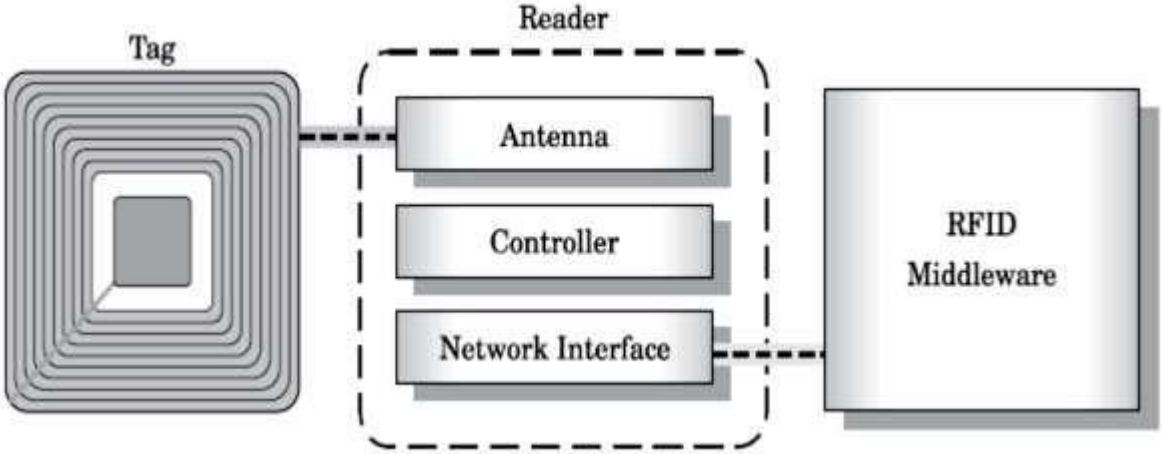
- Capacitive coupling, another form of close coupling that works best with the tag inserted intothe reader, is also usually used for smart cards.
- Capacitive coupling dispenses with antennas and replaces them with electrodes.
- The reader and tag each have conductive patches that together form a capacitor when held exactly parallel to each other without touching.



UNIT-III
READERS, PRINTERS AND READER PROTOCOLS

Physical Components of an RFID Reader

- Since the reader communicates with tags using RF, any RFID reader must have one or more antennas. Because a reader must communicate with some other device or server, the reader must also have a network interface of some sort.
- Examples of common network interfaces are the serial Universal Asynchronous Receiver/Transmitters (UARTs) for RS 232 or RS 485 communications and the RJ45 jack for 10BaseT or 100BaseT Ethernet cables; some readers even have Bluetooth or wireless Ethernet communications built in.
- Finally, to implement the communications protocols and control the transmitter, each reader must have either a microcontroller or a microcomputer.



Antenna Subsystem

- Although the antennas themselves are simple in concept, engineers work constantly to get better reception at lower power and to adapt the antennas to special circumstances.
- Some readers have only one or two antennas, packaged with the readers themselves; other readers may be able to manage many antennas at remote locations.
- The primary limitation on the number of antennas a reader can control is the signal loss on the cable connecting the transmitter and receiver in the reader to the antennas.

Controller

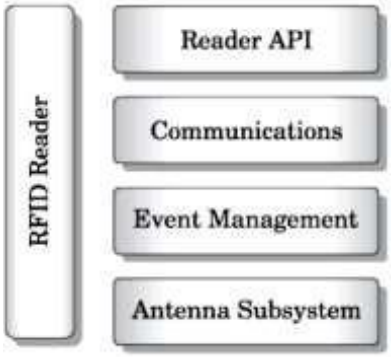
- The computing device that controls a reader can vary in complexity from a simple state machine on a chip, which might be used for a tiny embedded reader on a telephone or PDA, to a complete microcomputer system capable of running a server operating system as well as end user applications and accumulating a large amount of data on an internal hard disk.
- The controller is responsible for controlling the reader side of the tag protocol as well as determining when information read from a tag constitutes an event to send to the network.
- The reader controller is also responsible for managing the reader's end of the reader protocol.

Network Interface

- Reading tags and recognizing events wouldn't be much use if the reader never told anyone about those events. Readers communicate with the network and other devices through a variety of interfaces.
- Historically, most RFID readers have serial interfaces using RS232 or RS 422 (point to point, twisted pair) or RS 485 (addressable, twisted pair).

Logical Components of an RFID Reader

Within an RFID reader's controller, we can imagine four separate subsystems that handle different responsibilities.



Reader API

- Each reader presents an application programming interface (API) that allows other applications to request tag inventories, monitor the health of the reader, or control configuration settings such as power levels and the current time.
- This component is mostly concerned with creating messages to send to the RFID middleware and parsing any messages received from the middleware.
- The API may be synchronous or asynchronous.

Communications

- The communications subsystem handles the details of communicating over whatever transport protocol the reader may use to communicate with the middleware.
- This is the component that implements Bluetooth, Ethernet, or a proprietary protocol for sending and receiving the messages that make up the API.

Event Management

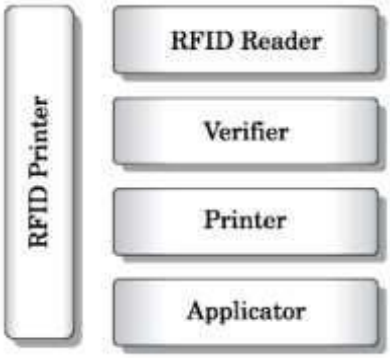
- When a reader "sees" a tag, we call this an "observation." An observation that differs from previous observations is called an "event." Separating out these events is called "event filtering."
- The event management subsystem defines what kinds of observations are considered events and which events are considered interesting enough to put in a report or send immediately to an external application on the network.
- As readers become smarter, they will be able to apply more complex processing at this level to reduce network traffic.
- Essentially, some parts of the event manager component of the middleware will naturally migrate to merge with the event management component of the reader.

Antenna Subsystem

- The antenna subsystem consists of the interfaces and logic that enable the RFID readers to interrogate the RFID tags and control the physical antennas.

Parts of RFID Printer and Applicator:

- RFID printers are devices that both encode tags and print to the paper labels that house the tags. Remember that a reader can also "write" to a tag that allows writes, so the primary difference between an RFID reader and an RFID printer has nothing to do with the capability to encode tags; the difference has only to do with the laser or inkjet printer component of the RFID printer.
- For low-volume applications, an operator may manually apply smart labels, but high-volume applications require a so-called "print and apply" device.



a. Reader

- RFID printer vendors, with some exceptions, are not usually reader vendors, so print and apply devices typically incorporate readers from one of the usual vendors.
- The print and apply device usually wraps the reader API in a print and apply API, but then offers a way to get to the reader API if necessary.

b. Printer

- The printer on an RFID print and apply device is no different than the printer on any other bar code printer, but bar code printers are very different than the office inkjet.
- These printers typically run a tape or "tag web" through spools so that they can print to one side and then flip the tag into position for application.
- All of these printers have the ability to print tags according to templates that describe the proper tag layout.

c. Verifier

- Print and apply devices usually include an RFID verification step as well as a bar code verification step.
- Typically, the RFID verification is performed by the same reader that wrote the tag and the bar code verification is performed by an optical scanner located just past the printer.

d. Applicator

- The applicator is one of the most important distinguishing characteristics of print and applies systems.
- Wipe-style applicators and tamp pad–style applicators have differing strengths and weaknesses.
- A wipe-style applicator has fewer moving parts and does not require an air compressor, but it can jam if a tag curls incorrectly.

- Tamp pad–style applicators are more reliable and arguably faster, but as mentioned earlier, many require an air compressor to drive the pneumatic ram.
- Electric solenoids can also be used to drive tamp pads, but the magnetic field generated by the coil may be a problem in RFID applications.

Layout for Readers and Antennas:

- A reader and its antennas must be installed to be of any use. Since with RFID we are attempting to sense qualities of the physical world - in this case, the presence or absence of particular items - the physical world dictates the specifics of any installation. For this reason, every sensor installation is different.
- The possible variations are infinite, but examining a few archetypal applications of RFID can help you to understand the broad categories of installation.

Portals

- The word "portal" means doorway or entrance, and an RFID portal is an arrangement of antennas and readers designed to recognize tagged items entering or leaving through a doorway. This is a common setup for warehouses, where items arrive and leave through loading docks.
- It can also be useful for items moving between sections of a factory, where tagged items might travel through doors (for example, moving from storage to the assembly floor).
- Portals may also be mobile; in these applications, the reader and antennas are built into a framework on wheels that can be pushed into a truck or down an aisle. This is useful for loading and unloading and for material tracking.

Tunnels

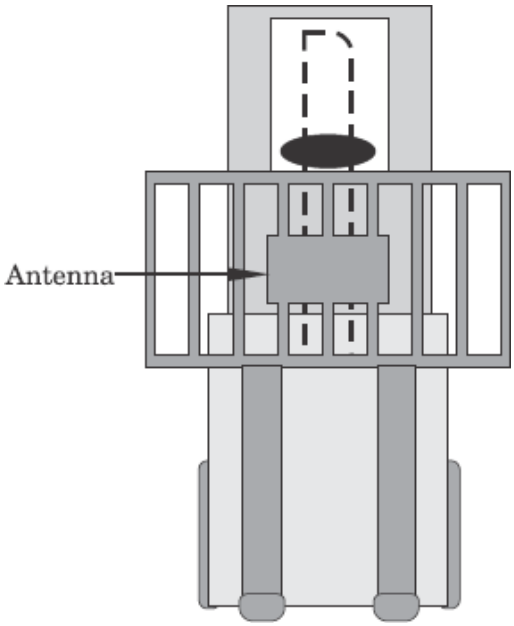
- A tunnel is an enclosure, usually over a conveyor belt, in which the antennas (and sometimes even the reader) may be housed.
- A tunnel is like a small portal, with the advantage that a tunnel may also include RF shielding, which absorbs reflected or misdirected RF energy that might interfere with other readers and antennas nearby.
- This can be useful for assembly lines or packaging conveyors where the reader identifies the station through which an item is currently passing on the conveyor.

Handhelds

- A handheld reader with integrated antenna, controller, and communications can allow personnel to scan tagged items in situations where it is inconvenient or impossible to move the items to a reader.
- The use of handheld RFID readers is very similar to that of handheld bar code readers.

Forklift Readers

- Forklifts, too, may carry RFID readers, for the same reason that a person might carry a handheld reader.
- Forklift manufacturers are beginning to offer RFID readers as part of the optional equipment on their products, just as they have offered bar code readers and operator terminals in the past.
- One pitfall of adding forklift readers in-place is the liability and regulatory concerns of adding equipment to these vehicles.
- Even prototype implementations must take care to include the manufacturer or even delegate installation of the hardware to the forklift manufacturer.



Smart Shelves

- One of the most talked about but least common applications of RFID is the smart shelf.
- Smart shelves are shelving units with antennas incorporated into them in such a way that readers can recognize the arrival and departure of items from the shelves, or read all the items on the shelves on demand.

Vendor Protocols:

- Different RFID reader vendors have created significantly different reader protocols, but all of them perform the same basic functions. In the following sections, we examine a simple "hello world" RFID application using reader protocols from two of the leading reader manufacturers, Alien and Symbol.
- The Symbol reader is the Matrics AR-400 (Symbol acquired another leading reader vendor, Matrics, in 2004).

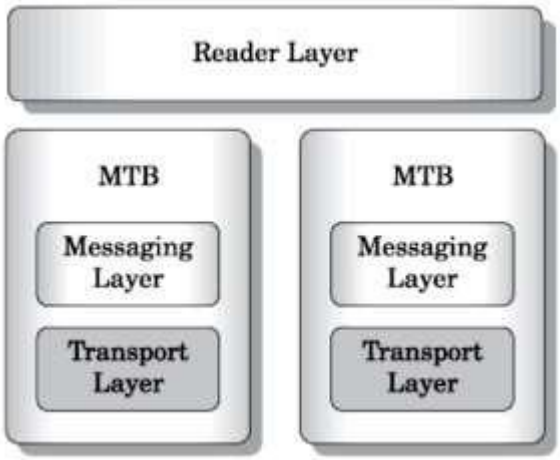
a. Alien

- Alien Technology prefers to use the terms Interactive mode and Autonomous mode for these two types of communication rather than synchronous and asynchronous, but the respective steps performed by the reader and host are the same.
- The Alien reader accepts commands over a serial port or through a telnet session via the Transmission Control Protocol (TCP).
- Some configuration commands may also be supplied through a web interface using HTTP GET and POST commands (implemented as a web GUI).
- Alien supports notifications of observations or alerts by email (via the Simple Mail Transfer Protocol), over a TCP socket, or over the serial port, using several configurable formats for the information.
- However, writing a complete middleware implementation becomes much more challenging when we consider the need to monitor and manage the reader, configure replacement readers, and push software upgrades to the reader as updates come out.
- Alien provides a management console for its readers, but it cannot manage readers from other vendors or other types of sensors.

b. Symbol

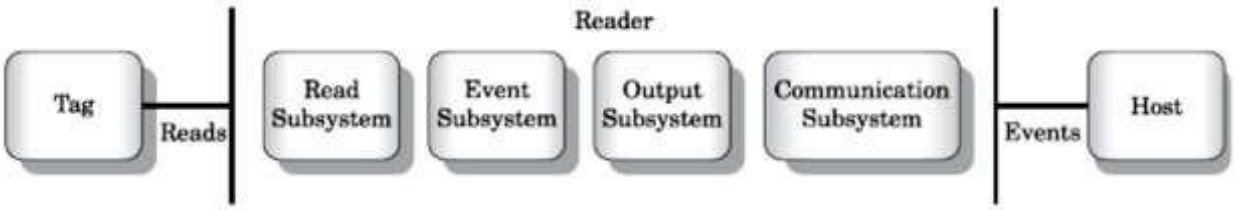
- The AR-400 from Symbol Technologies accepts commands as XML over HTTP or as XML over a TCP socket or a serial port; it also supports a vendor-specific byte stream protocol over a TCP or serial connection.
- Notifications may be configured as synchronous, which Symbol calls "Query mode," or synchronous, called "Publish/Subscribe mode" in the documentation.
- The AR-400 supports the Simple Network Management Protocol (SNMP) for alerts and configuration and can also accept configuration as XML or byte stream commands. It supports Ethernet and serial transports.
- The AR-400 has an embedded HTTP server, which provides the Reader Administration Console.

Various layers of the EPCglobal Reader Protocol:



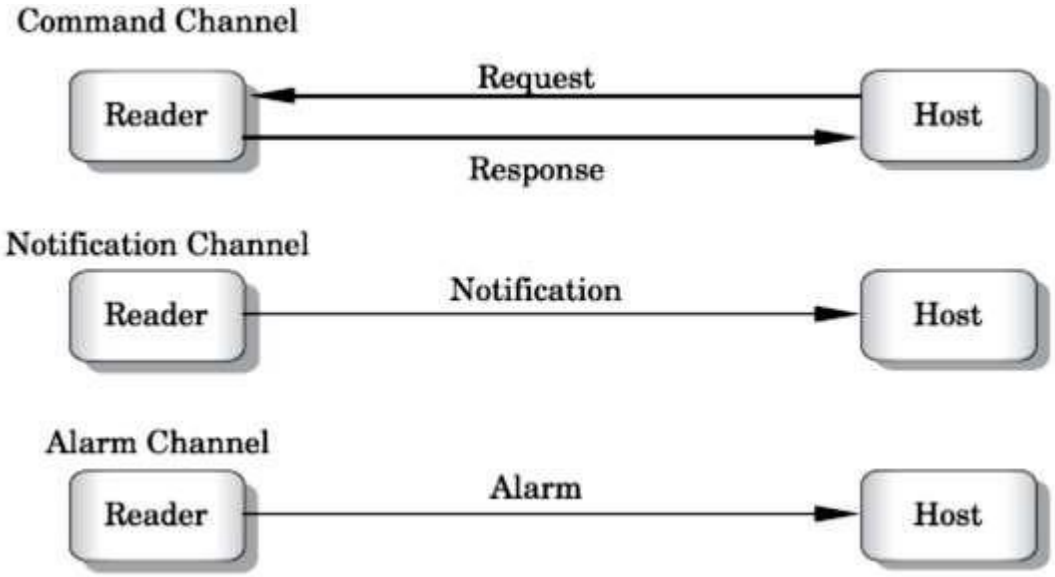
Reader layer

- The Reader layer defines the allowable content and format of messages sent between the reader and host. This layer comprises the Open System Interconnection (OSI) Presentation and Application layers.
- The protocol specifically allows this layer to make use of multiple MTBs, although a particular instance will use only one MTB at a time. Also, regardless of the MTB, the reader can hold a conversation with only one host at a time.



Messaging layer

- This layer lives on top of the Transport layer and is responsible for managing connections and security and for packaging the host commands and reader responses and notifications so they can pass back and forth on the Transport layer.
- Any encryption, authentication, or session management occurs in this layer. This layer describes how a conversation between the reader and host starts and stops and defines the shape of the frame, or the envelope in which messages pass back and forth. This layer is logically identical to the OSI Session layer.



Transport layer

- This is the lowest layer, and it describes either the services provided by an OS or the hardware needed to support networking. It corresponds roughly to the OSI Physical, Data Link, and Network layers.
- The Transport layer is the physical connection and the networking connection between the reader and host - for instance, TCP over Ethernet using Cat 5 cables, an RS485 network over twisted pair, or a wireless Bluetooth implementation.

Different parts of a Reader Protocol:

Alert

- An alert is a message from a reader to a host indicating a change in reader health or containing a scheduled update of reader health information.

Command

- A command is a message from a host to a reader that causes a change in state in the reader or a reaction from the reader.

Host

- A host is an application or middleware component that communicates with readers.

Observation

- An observation is a record of some value somewhere at some time for example, the exact temperature inside a refrigeration unit at a particular point in time, or the appearance of ID tag 42 at dock door 5 at 16:22:32 on July 23, 2005.

Reader

- A reader is a sensor that communicates with tags to observe identities and then communicates these observations to a host.

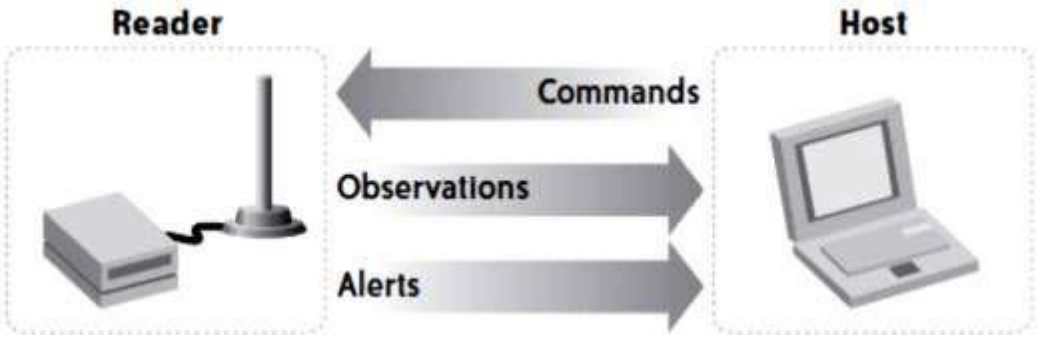
Transport

- A transport is a communications mechanism used by readers and hosts to communicate with each other.

Trigger

- A trigger is some criterion, such as time of day, that will cause some activity to occur.
- An example might be a timed read trigger that causes a reader to attempt to read any tags present every 12 minutes.

- With these terms described, we can define a reader protocol as a set of formal rules defining how one or more hosts and one or more readers may communicate commands, observations, and alerts over a transport.
- Any reader protocol must deal with three major types of communication: commands passing from the host to the reader, observations passing from the reader to the host, and alerts passing from the reader to the host.



Commands

- A host sends commands to a reader to cause some reaction from the reader or to change the state of the reader in some way.
- We can broadly divide the commands a host sends to a reader into three categories:

a. Configuration commands

- These commands are for the setup and configuration of the reader.

b. Observation commands

- These commands cause the reader to read, write, or modify tag information immediately.

c. Trigger commands

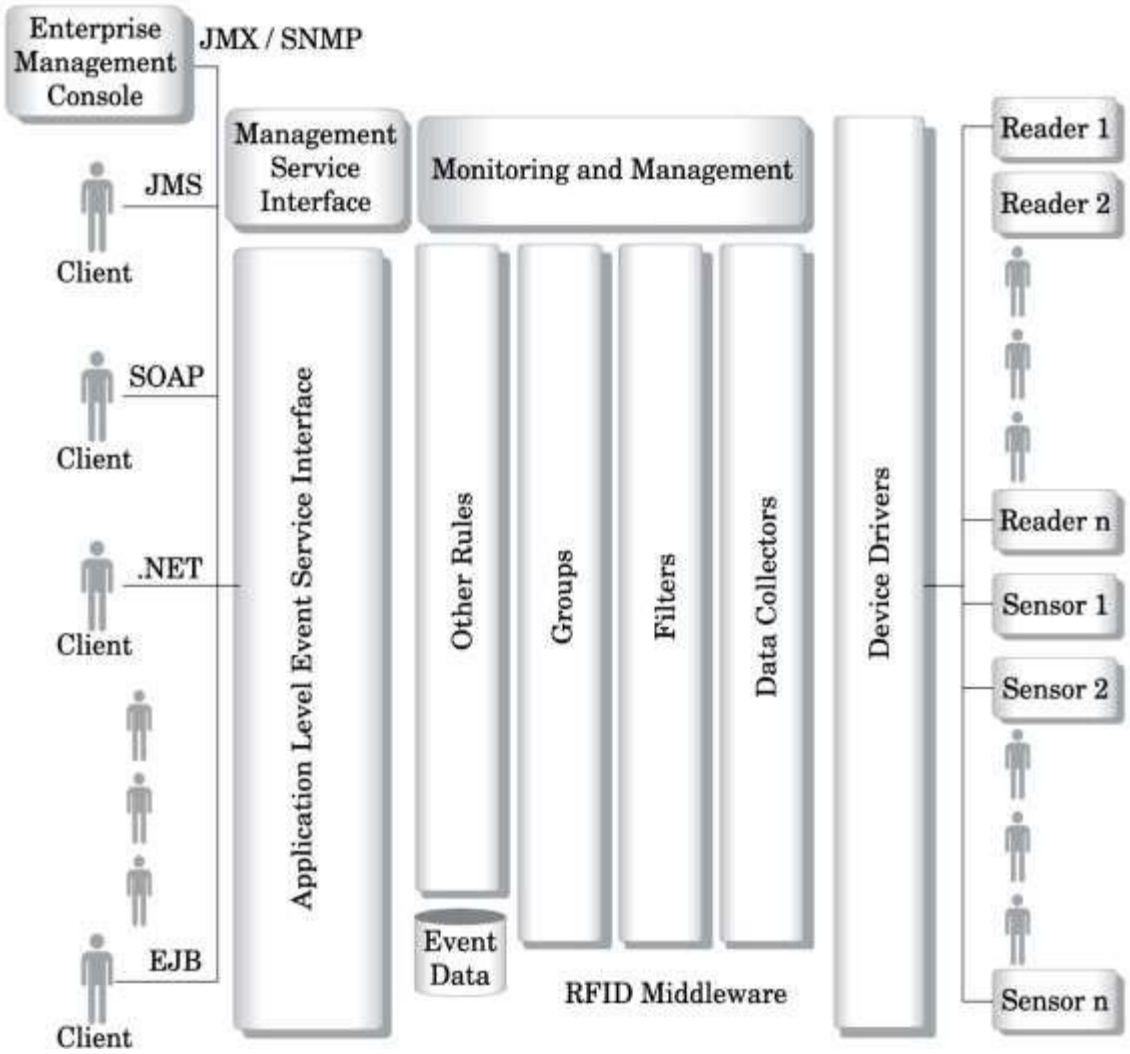
- These commands set triggers for events such as reads or notifications to occur.

UNIT-IV
MIDDLEWARE AND INFORMATION SERVICE

Motivations:

- There are three primary motivations behind using RFID middleware: to encapsulate the applications from device interfaces; to process the raw observations captured by the readers and sensors so that applications see only meaningful, high-level events, thereby lowering the volume of information that they need to process; and to provide an application level interface for managing readers and querying RFID observations.

Logical Architecture:



- RFID and other remote sensing technologies provide a level of automation that was not previously possible with labeling technologies such as bar codes that needed human intervention.

- However, this level of automation requires that the readers and sensors be monitored and managed remotely.
- A middleware solution that operates at the edges is best suited to monitoring and managing edge devices.
- Thus, in addition to the three functions described above, an RFID middleware solution should also provide, or at least integrate with, a management and monitoring interface.

Key concepts and Terminologies involved in Application Level Events Specification:

Event originators

- An event originator is any device that captures the presence of an RFID tag or any other observation from the physical world.
- RFID readers and sensors are examples of event originators.

Read cycles

- To understand event management, it is important to understand how events are originated and passed from the Readers to the ALE server and then on to the clients of the ALE server.
- A Reader can scan for RFID tags or other physical observations at a set frequency or on demand. When scanning is done at a set frequency, each scan is called a read cycle.
- A read cycle is a unit of interaction with a Reader. Each read cycle results in a Reader returning a set of RFID observations. In the context of the ALE specification, each observation is an Electronic Product Code (EPC).
- The ALE specification does not restrict how Readers support read cycle timing. Read cycles can be based on time duration, data volume, other sensor inputs, and so on.

Event cycles

- Sometimes, an application may want information collected over several read cycles.
- For example, an inventory management system may want to get updates every hour on what inventory items should be ordered.
- This suggests that we need a higher-level abstraction that can group a series of read cycles. Here is where the concept of the event cycle comes in.
- An event cycle is a unit of interaction that a client uses with an ALE service.
- There is considerable flexibility in how event cycles map to read cycles.
- For instance, an event cycle can span multiple read cycles, and multiple event cycles can span a given set of read cycles.
- An event cycle can also span read cycles from multiple readers.
- Given this flexibility, it is important to note that a client sees an event cycle as a unit and does not see the individual read cycles that make up a given event cycle.

Interaction Models

- The interaction models supported by the ALE specification should come as no surprise to application developers: a client can either request services on demand (synchronous mode) or register for information to be sent to it when certain conditions are met (asynchronous mode).

Commercial RFID Middleware Platforms:

Sun Microsystems

- Sun Microsystems was one of the early entrants into the RFID market.
- Sun provides a Java-based RFID middleware platform called the Sun Java System RFID Software.
- Sun's RFID software is designed specifically to provide high levels of reliability and scalability for your EPC network, while also simplifying the task of integrating with multiple existing backend enterprise systems.
- Sun's RFID middleware is part of the Java Enterprise System (JES) and supports standards-based integration with leading enterprise integration servers, including the Sun Java Enterprise Integration Server.
- The four components of the project are the RFID Event Manager, the RFID Management Console, the RFID Information Server, and a software development kit (SDK) for creating adapters and standalone applications.

RFID Event Manager

- The RFID Event Manager is a Jini-based event management system that facilitates the capture, filtering, and eventual storage of EPC events generated by RFID readers connected to the network.
- Its main goals are to interface with RFID readers, gather EPC events, filter redundant information, and feed relevant events to the RFID Information Server or other ERP software for further processing.
- The RFID Event Manager Jini Services are managed through Rio, an open source container of Jini Service Beans.

RFID Management Console

- The RFID Management Console (MC) is a browser-based graphical interface used to manage and monitor the RFID Event Manager. It allows the user to view and modify the RFID reader attributes and components of the Event Manager, such as filters and connectors.
- The RFID MC is a Struts application that provides the same look and feel as the Sun Control Station.
- It provides the user with a table-based display of RFID readers and RFID Event Manager components.
- The attributes of these readers and components can be modified to alter the behavior of a running system.
- The RFID MC allows the user to group readers and display the status of a group for easy viewing.
- It also implements an alarm table of readers and their attributes.
- Each reader is displayed by group and is selectable.
- Once selected, the reader's attributes are displayed.

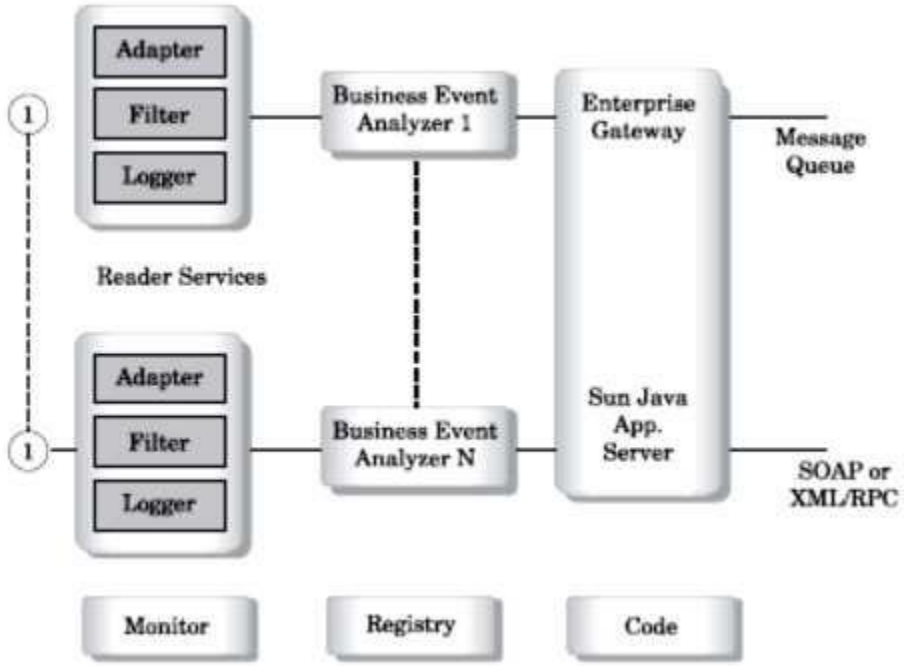
RFID Information Server

- The RFID Information Server (IS) is a J2EE application that serves as an interface for the capture and query of EPC-related data.
- EPC-related data can include tag observation data from Event Managers as well as information that maps EPCs to higher-level business data.

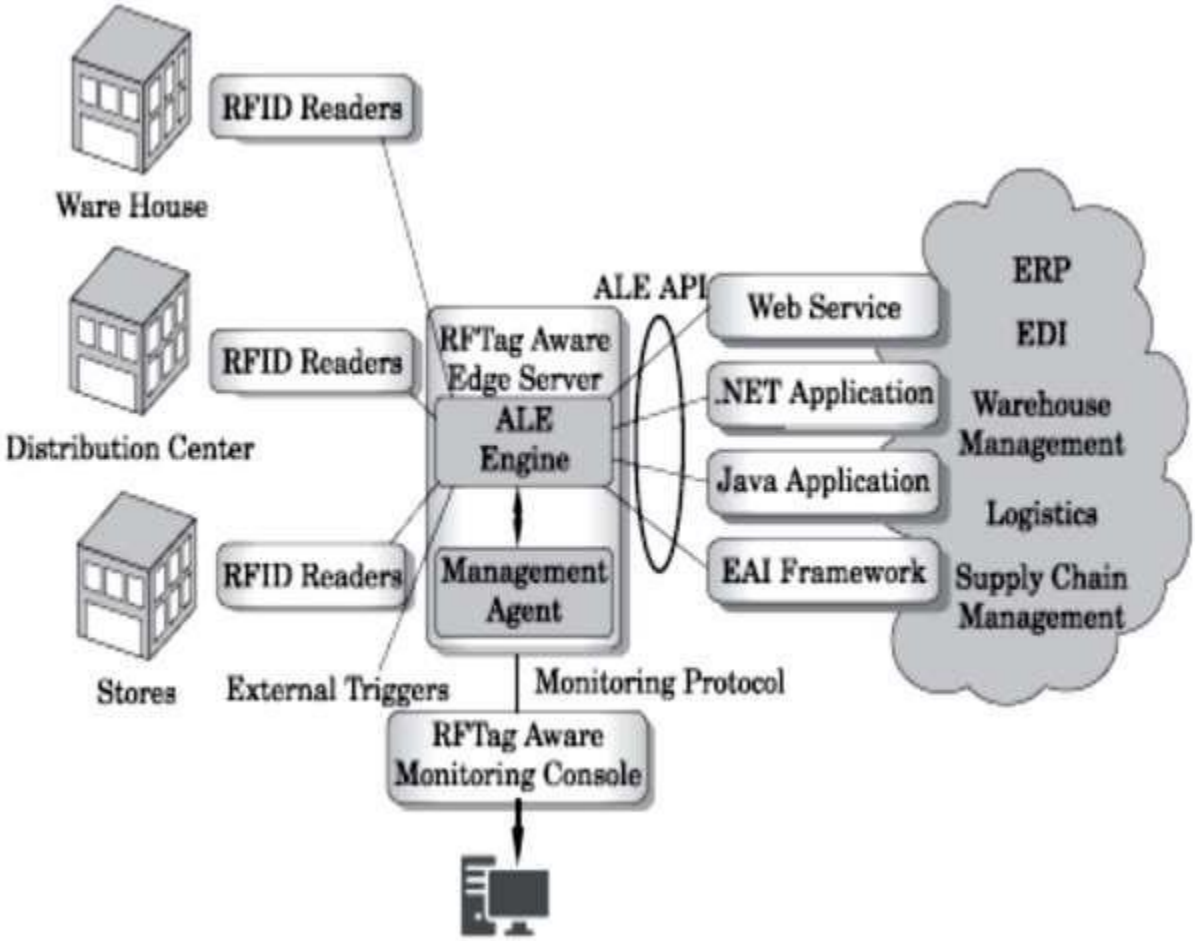
- The RFID IS is typically used to translate a set of low-level observations into higher-level business functions.

SDK

- The well-documented SDK allows developers to extend the product if they choose to create a custom application rather than using the components as they are shipped.



ConnecTerra / BEA



- ConnecTerra's primary product, RFTagAware, is a software infrastructure platform for the development of device applications and RFID solutions.
- ConnecTerra was one of the first companies to implement a middleware solution based on the Application Level Events (ALE) standard. ConnecTerra is very active in the EPCglobal standards groups, and its founder and CTO Dr. Ken Traub was the lead author for the Application Level Events specification.
- With RFTagAware, devices such as RFID readers are abstracted, similar to how a database is abstracted by SQL.
- Users simply describe the events they are interested in, much like a database query, and subscribe to the results that are produced based on the RFID activity.
- The RFTagAware Edge Server, a piece of software deployed on or near a device, processes the raw tag information and, based on any number of outstanding queries (known as Event Cycle Specifications), delivers qualifying results to any number of subscribing applications.
- Queries can be added, changed, or removed independently in real time without affecting the queries or results used by other applications.
- The Edge Server takes care of optimizing the use of the hardware, ensuring device utilization is optimized for the collection of outstanding queries.

RFID Data:

Event Data

- Event data is tied to a specific moment in time and communicates the whereabouts of an RFID-tagged asset as it moves through a supply chain.
- In essence, RFID event data is made up of observations of the existence of something at some place at some time.

Identity

- Any RFID observation must communicate some identity.
- Those were the same kinds of identities we expect here for example, an SGTIN, a GRAI, or an SSCC.

Location

- Location also ties into some of our earlier discussions.
- A particular location can be recorded as a GLN, but it probably corresponds physically to a specific source antenna.

Time

- Time is one of those recurring problems in designing any system.

Master Data

- In addition to these core attributes, which are collected as event data, you may have a host of other reference or contextual information about the item being tracked that is meaningful in your business context.
- Master data provides supporting contextual or reference information about the event data.
- This could include a description of the product referenced by the EPC, information about its manufacturer, or other kinds of information such as details about the physical location where the event was captured.
- Generally, your RFID master data will not grow at the same pace as the event data.

Data Volume

- The impact of RFID data on your supply chain and other asset tracking systems will depend on what data you collect using your RFID infrastructure, how often you collect it, how much of it you collect, and what you plan to do with it.
- Depending on the size of your RFID infrastructure and the granularity at which you need to track your assets, RFID data volume can potentially overwhelm your networks and storage.

Data Storage

- The database back at headquarters will be accumulating the observations and storing them for some period of time.

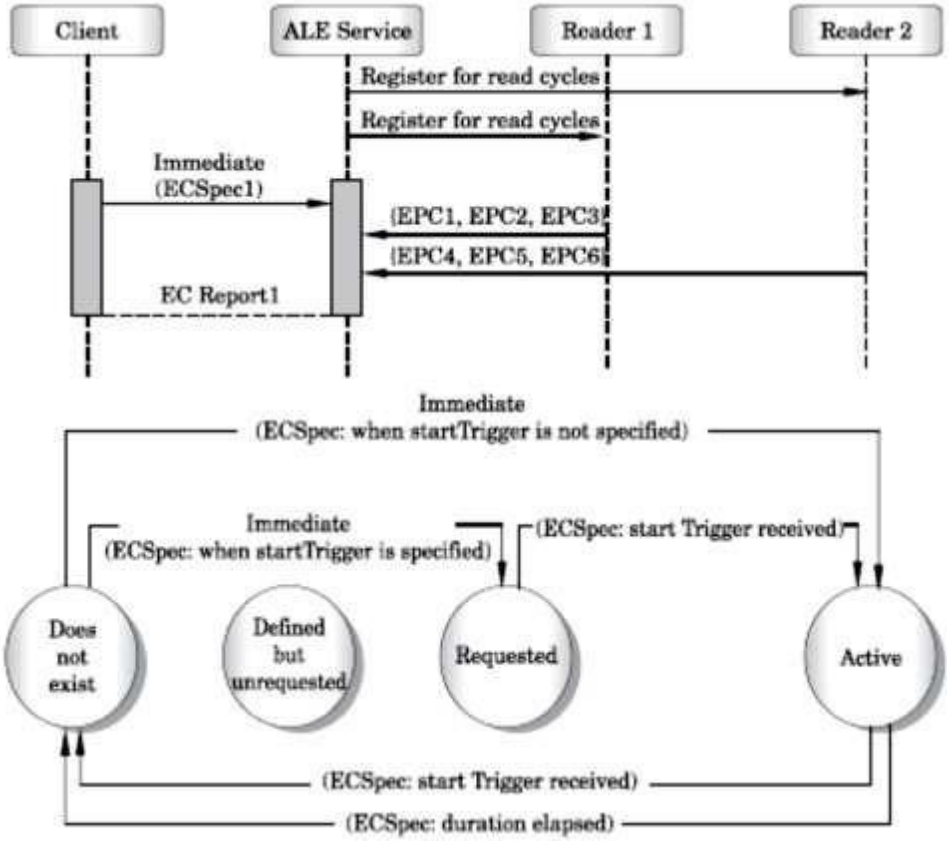
ALE specification:

Synchronous mode

- The ALE specification calls this mode of interaction "immediate."

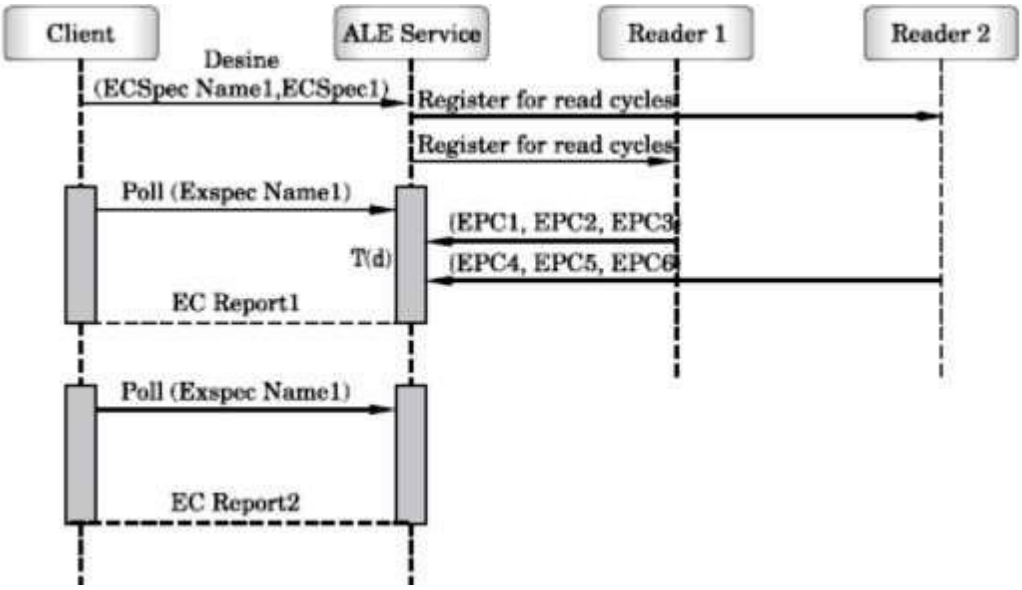
Immediate mode

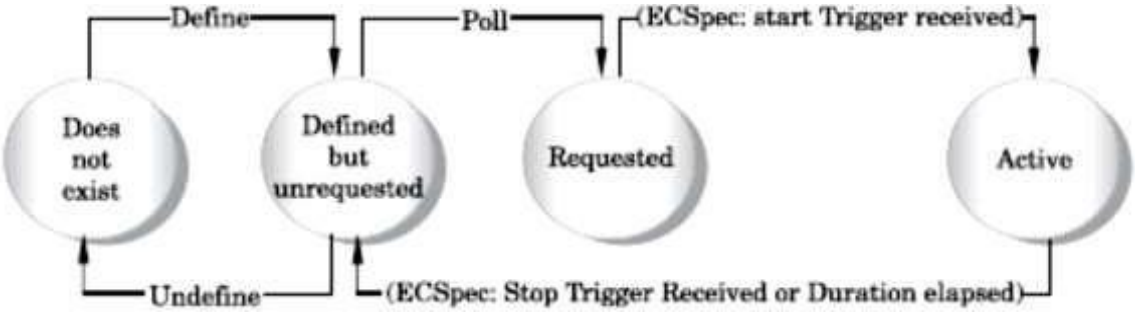
- Immediate mode is one of the two synchronous methods available to access an ALE service.
- The client first creates and configures an ECSpec, and then invokes the "immediate" service of the ALE server.



Polling mode

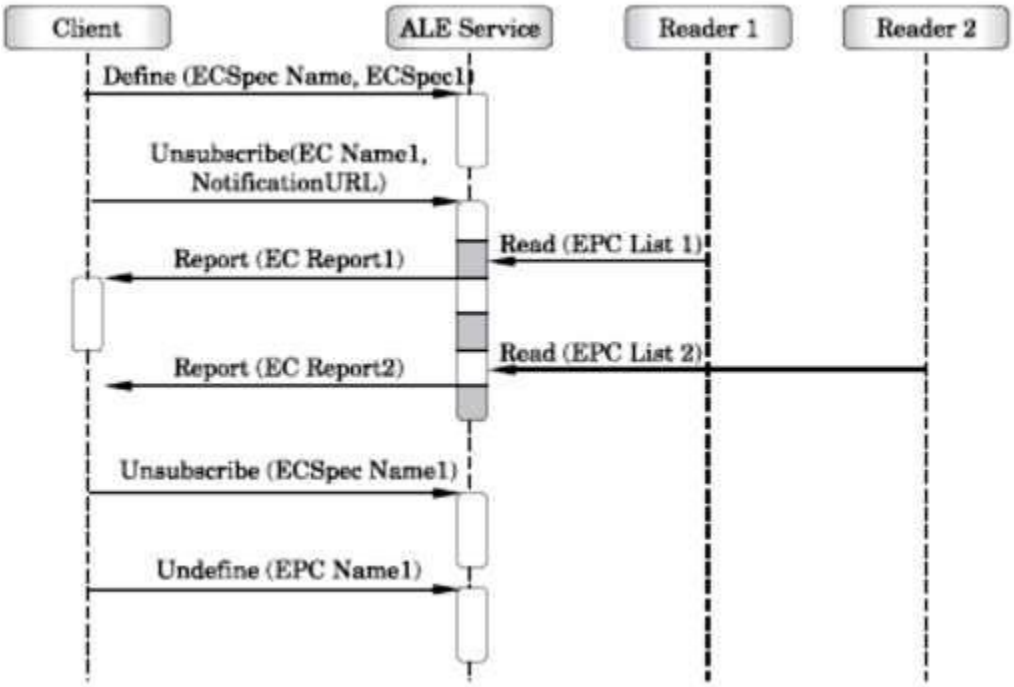
- A client that wants to get regularly scheduled updates rather than one time reports, on EPC event data would use the ALE server's polling interface.
- Polling is executed synchronously.
- To some extent, it is similar to subscribing and then unsubscribing after an event cycle is generated.

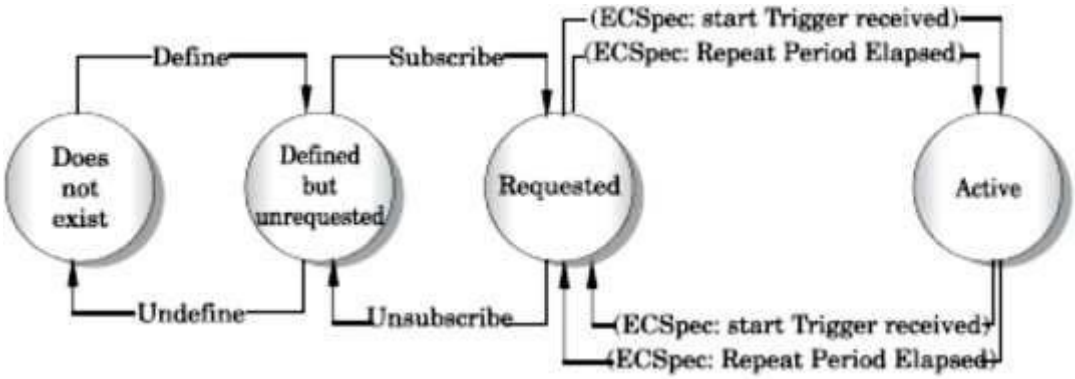




Asynchronous mode

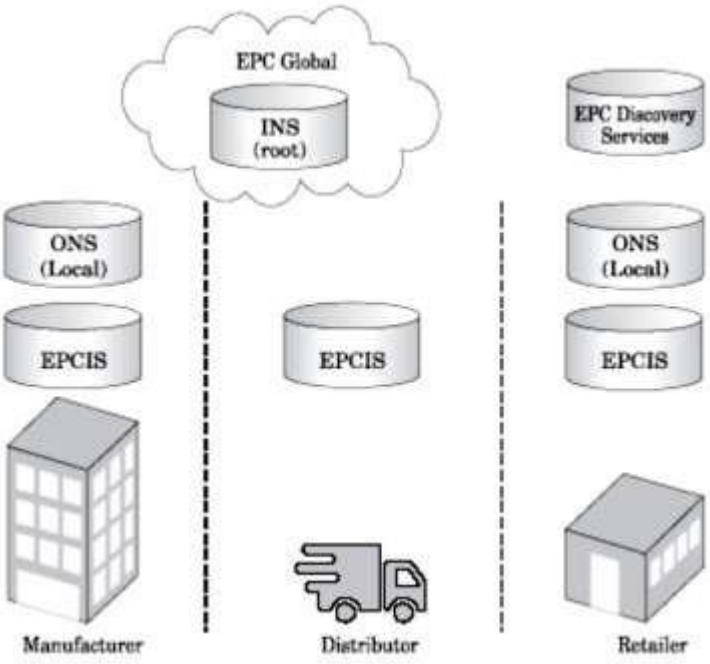
- The asynchronous ALE service interaction mode uses the classic publish/ subscribe mechanism used in other asynchronous messaging architectures.





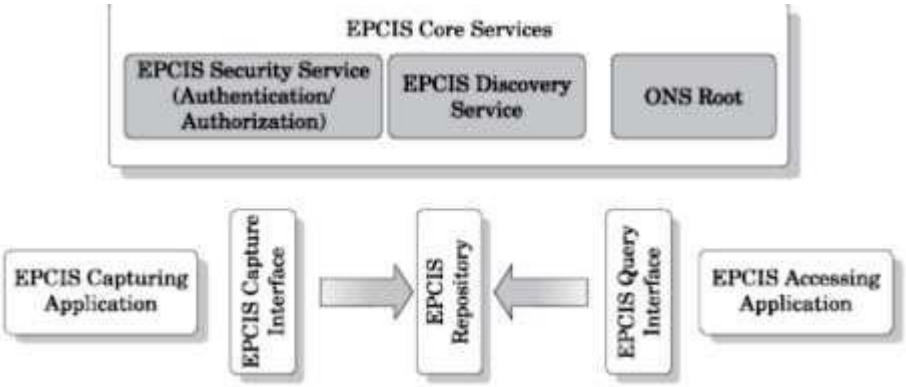
EPCglobal Network & the components of EPC information services:

- EPCglobal envisions a network of EPC-enabled data services that is used by trading partners to enable near-real-time tracking information on items in their supply chains.
- This vision is termed the EPCglobal Network.
- The EPCglobal Network introduces a few dedicated components, such as the Object Naming Service (ONS) and the EPC Information Services (EPCIS), that you may or may not need for your applications.
- However, chances are that you will end up using or developing similar components to meet your requirements.
- The EPCglobal Network aims to provide real-time data about individual items as they move through a global supply chain.
- Centeredaround the EPC and RFID technologies, and building on the existing Internet infrastructure, the EPCglobal Network offers the potential for increased efficiency and accuracy in tracking products between trading partners.
- The EPCglobal Network is made up of five principal services:
    - ❖ Assigning unique identities
    - ❖ Detecting and identifying items
    - ❖ Collecting and filtering events
    - ❖ Storing and querying events
    - ❖ Locating EPC information



The EPC Information Services

- The EPCIS is an upcoming EPCglobal standard whose goal is to enable disparate applications to leverage EPC data via EPC-related data sharing, both within and across enterprises.
- The EPCIS defines a standard interface for capturing and sharing EPC-related data.
- It should be noted that the EPCIS focuses only on the service interface and semantics of EPC-related data, such as location information that gets registered as products move through the supply chain.
- Vendors are provided the flexibility to compete on implementations and add-on functionality.
- EPC observations are captured using the EPC Capture interface, and they are queried using the EPC Query interface.
- Additionally, the EPCIS provides a common model for location information and other important data.



UNIT-V
FLEXIBLE SENSORS

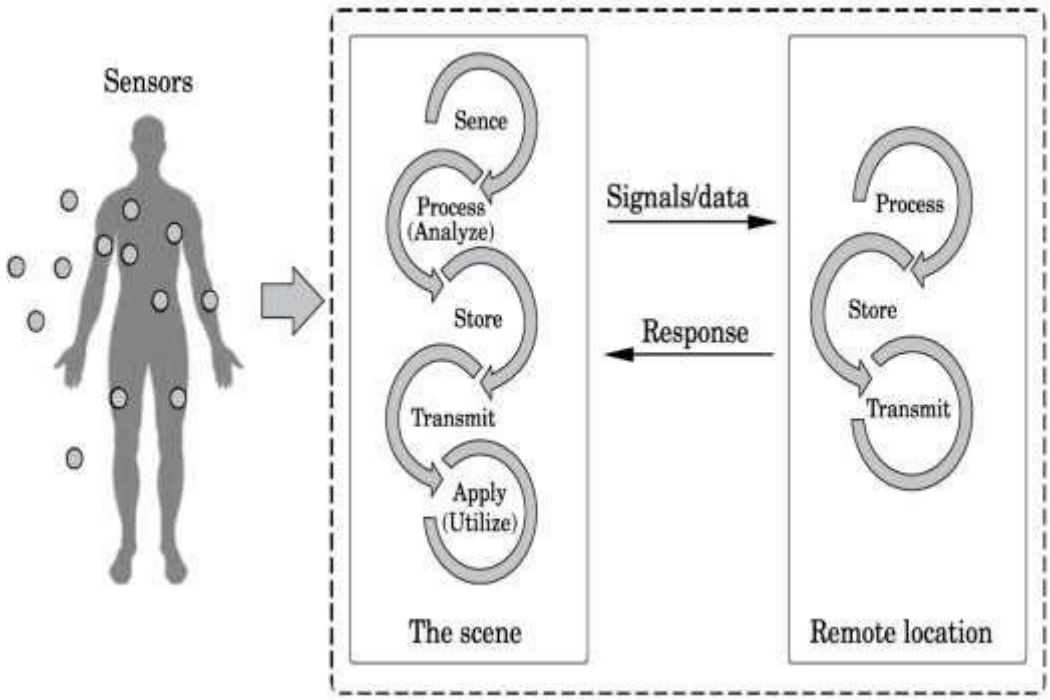
World of Wearables (WOW):

- In today's digital world the term "wearable" has a new meaning.
- Today it brings up images of accessories such as a smart watch on a business executive's wrist, a head-mounted display worn by an immersive gamer, a tiny sensor on a cyclist's helmet, or a smart garment a runner uses to track and monitor her steps.

- In recent years, the dimensions of fashion and protection typically associated with the traditional wearability of clothing have expanded to include "functionality" on the go.
- This functionality can essentially be characterized as mobile information processing – whether it is the executive checking e-mail, the gamer shooting at a target that is also being simultaneously chased by a fellow gamer on the other side of the world, the trainer ensuring that the rider is maintaining proper posture on the curve, or the runner tracking her workout and caloric intake for the day.
- Just as clothing can be personalized and customized for each person (depending on the physical dimensions, taste, and style preferences) and/or occasion (business, evening, casual, home, and hiking), the new wearable too can be configured for personalized mobile information processing for specific applications such as immersive gaming, fitness, public safety, entertainment, healthcare, etc. In short, the world of wearables (WOW) is transforming our lives.

Role of wearables:

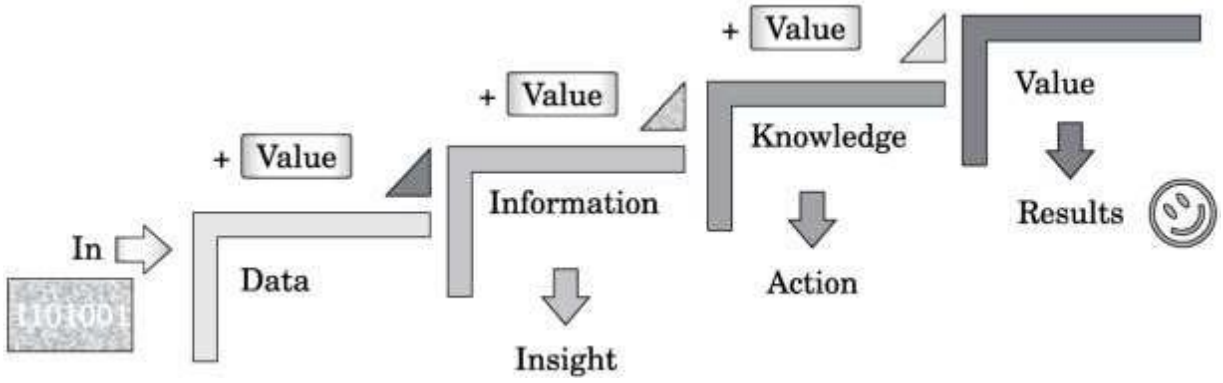
- Fundamentally, wearables can perform the following basic functions or unit operations in each of the scenarios
    - Sense
    - Process (Analyze)
    - Store
    - Transmit
    - Apply (Utilize)



- Simultaneously, it may be transmitted to a remote location for confirmatory testing and the results – along with any appropriate response (i.e., put on a gas mask) – can be communicated to the user in real-time to potentially save a life.
- This same philosophy can also be used by an avid gamer who might change his strategy depending on what "weapons" are available to him and how his opponents are performing.
- Each of these scenarios requires personalized mobile information processing, which can transform the sensory data into information and then to knowledge that will be of value to the individual responding to the situation.
- While wearables are being used in many fields.
- Wearables provide an unobtrusive way to longitudinally monitor an individual not just during the day but, over the individual's lifetime.
- Such an expansive view of the individual will be valuable in detecting changes over time and help in early detection of problems and diseases leading to preemptive care and hence, a better quality of life.
- Inferring the potential of wearables in other application domains should be straight forward and can be accomplished by instantiating the fundamental principles and concepts.

Data-information-knowledge-value paradigm

- In triage, the nurse documents the vital signs gathered using instruments (e.g., thermometer, blood pressure monitor, EKG machine) that convert the raw signals (the data) from the body into meaningful information (temperature, diastolic/systolic pressure, heart rate, and electrocardiogram) and thus add value as shown in the figure.
- When the physician processes this information, he or she gains insight into the potential condition of the patient.



- The physician adds value by drawing upon the knowledge,expertise and experience accumulated over time to come up with a diagnosis and a plan of action or treatment.
- This course of treatment is in the form of medication and other interventions is the value derived by (or delivered to) the patient resulting in the curing of the illness.
- Thus, the raw data gathered by the instruments is valuable only when it is properly transformed and harnessed to benefit the individual.
- For this transformation to occur seamlessly, there is a need for an information/knowledge processing ecosystem.

Big data analysis

- The emerging concept of big data Park and Jayaraman discussed the role of wearables in relationship to "big data".
- Big data refers to large amounts and varieties of fast-moving data from individuals and groups that can be processed, analyzed, and integrated over time to create significant value by revealing insights into human behavior and activities.
- According to McKinsey, if the US healthcare system could use "big data creatively and effectively to drive efficiency and quality," the estimated reduction in healthcare spending could be between $300 billion and $450 billion per year, which would be a sizeable chunk of the annual healthcare spending costs.
- Let's consider one such example in healthcare, an application domain in which wearables are being increasingly deployed.

Medical loss ratio and wearables

- The Patient Protection and Affordable Care Act of 2010 requires health insurance companies to spend at least 80% to 85% of premiums collected on providing medical care.
- Known as the Medical Loss Ratio (MLR), the objective behind this provision is to bring down the overhead costs of providing medical care and limit it to 20% for individual and small-group coverage and 15% for large-group coverage.
- With the increasing shift from volume-based to value-based reimbursement for services rendered, healthcare providers are incentivized to provide holistic care to patients by closely monitoring them to ensure compliance with medication and promoting healthier lifestyles.
- Wearables enable this remote health monitoring of patients.
- The health data can be wirelessly sent to the physician's office by the wearable, negating the need for office visits.
- Consequently, the cost of care decreases.
- Moreover, the ability to continuously track patients' health can help identify any potential problems through preventive interventions and thus enhance the quality of care while eliminating unnecessary procedures since the cost of prevention is significantly less than the cost of treatment.

- The resulting higher quality of care at lower costs would also contribute to better operating efficiencies and lower overhead costs for insurance companies since their resources can be better spent on actually providing care and not on measures to ensure that a high quality of care is being provided.

Challenges and opportunities for wearables:

- The success of any innovative product in the marketplace depends on:
  - Its effectiveness in successfully understanding the user's needs and meeting them
  - Its compatibility with or similarity to existing products or solutions
  - The extent of behavioral change needed to use the new product
  - The reduction in the cost of current solutions or technologies it aims to supplant
  - The improvement in the quality of service (or performance)
  - The enhancement of the user's convenience
- The innovation should provide a tangible advantage to the user and it should be consistent and compatible with the user's values, beliefs, and needs.
- Much innovative technology has not been a marketplace success for one or more of these aforementioned reasons.
- For example, Apple's Newton, the first handheld device did not make it in the market but spawned the highly successful Palm Pilot and generations of personal digital assistants because the latter addressed many of the issues that plagued the Newton.
- In the process, they spawned the ongoing innovation in tablets. Thus, factors related to the diffusion of innovation must be considered in addition to the technical and business challenges to ensure the successful transition of wearables from the laboratory to the highly competitive market place.
- A roadmap analyzing the technical, business, and public policy issues, including the need for a "killer app" to influence the adoption and acceptance of wearables, has been proposed.

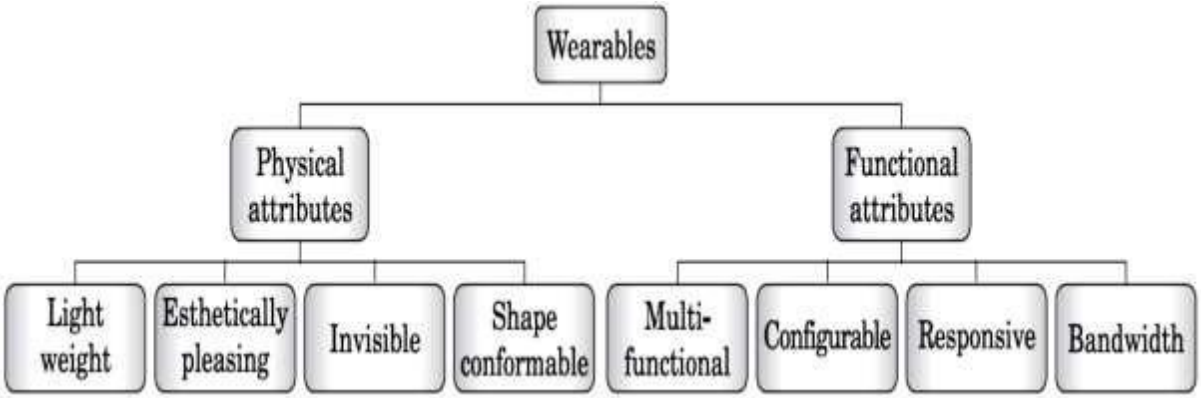
Technical challenges

- The success of wearables depends on the ability to connect them seamlessly in a body worn network. This means the meta-wearable framework must have the ability to route the signals and power between desired points in the structure.
- In the event of damage to the data buses in the meta-wearable framework, the "failure" in the network must be recognized and alternate "data paths" must be established in the fabric to maintain the integrity of the network by taking advantage of the redundant data buses in the fabric.
- Currently, the so-called "t-connectors" and button snaps are being used for connecting sensors and processors to the meta-wearable.
- Many of the wearables, especially those used for health monitoring applications and immersive gaming, are prone to motion artifacts, which can potentially affect the integrity of the results.
- While currently available conductive fibers can fulfill the basic requirements for the first generation of textile-based wearables, it is important to develop new materials that will have the conductivity of copper and the properties of textile fibers such as cotton, polyester, or nylon, and be available in commercial quantities.
- Today's wearables are powered by lithium-ion rechargeable batteries, which is another limiting factor in the adoption of the technologies due to the rigidity of the battery about the flexible nature of the wearables, a key desired attribute of wearables.
- The seamless integration of wearables in health care settings and for remote monitoring faces the challenge of ensuring compatibility with existing wireless technologies and established operational protocols in those settings.

- The supply chains for textiles/clothing and electronics are significantly different. Apparel manufacturing is a labor-intensive operation whereas electronics manufacturing is highly automated.
- Finally, the same wearable may be used in a range of environmental conditions – indoors to outdoors – which may include disaster zones involving high temperatures (e.g., fire) and hazardous materials. Therefore, they should be designed to function effectively and seamlessly in a wide range of ambient environments.

Attributes of Wearables

- From a physical standpoint, the wearable must be lightweight and the form factor should be variable to suit the wearer.
- For instance, if the form factor of the wearable to monitor the vital signs of an infant prone to sudden infant death syndrome prevents the infant from (physically) lying down properly, it could have significant negative implications.
- The same would apply to an avid gamer if the form factor interferes with her ability to play "naturally," the less likely that she would be to adopt or use the technology.
- Esthetics also plays a key role in the acceptance and use of any device or technology.
- This is especially important when the device is also seen by others i.e., the essence of fashion.
- Therefore, if the wearable on a user is likely to be visible to others, it should be esthetically pleasing and, optionally, even make a fashion statement while meeting its functionality.
- In fact, with wearables increasingly becoming an integral part of everyday lives, the sociological facets of the acceptance of wearables open up exciting avenues for research.
- Ideally, a wearable should become such an integral part of the wearer's clothing or accessories that it becomes a "natural" extension of the individual and "disappears" for all intents and purposes.
- It must have the flexibility to be shape conformable to suit the desired end-use; in short, it should behave like the human skin.
- The wearable must also have the multifunctional capability and be easily configurable for the desired end-use application.
- Wearables with single functionality (e.g., measuring just the heart rate) are useful, but in practical applications, more than one parameter is typically monitored; and, having multiple wearables one for each function or data stream would make the individual look like a hypothetical person and deter their use even if the multiple data streams could be effectively managed.
- The wearable's responsiveness is critical, especially when used for real-time data acquisition and control (e.g., monitoring a first responder in a smoke-filled scene).

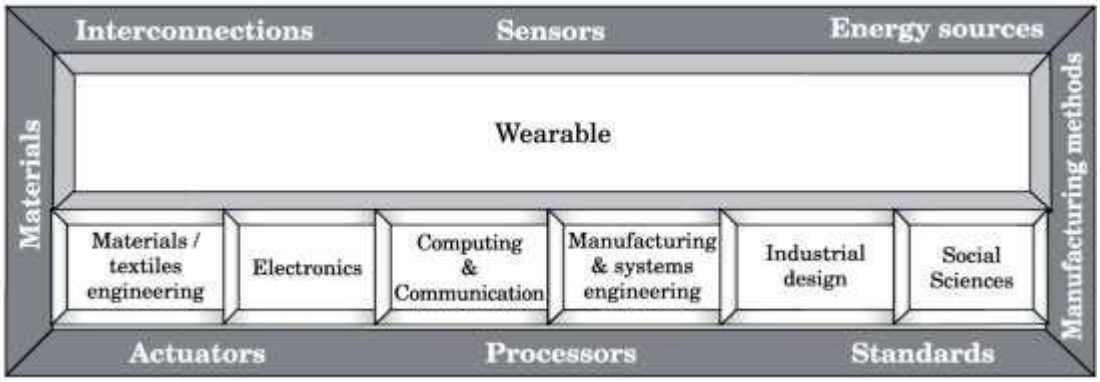


Textiles and Clothing : The Meta-wearable

- A critical need for extensive deployment of wearables for personalized mobile information processing is that they should not impose any additional social, psychological, or ergonomic burden on the individual.
- For instance, Google Glass significantly impacted social dynamics since the ones without this wearable device were not sure of what the wearer was doing with the device while being part of the conversation; consequently, it did not gain acceptance in the marketplace.
- What is, therefore, needed is an infrastructure or platform that will be unobtrusive, natural, and pervasive, and not adversely impact social interaction.
- Moreover, for many real-world applications, multiple parameters must be simultaneously acquired, processed, and used to develop an effec-tive response.
- This leads to the following requirements for creating and developing a useful wearable sensor system.
- Different types of sensors will be needed for various parameters to be monitored simultaneously; for instance, sensors to monitor the various vital signs (e.g., heart rate, body temperature, pulse oximetry, blood glucose level) are of different types. Likewise, for monitoring hazardous gases, another class of sensors (e.g., carbon monoxide detection) will be required. Accelerometers will be required to continuously monitor the posture of the gamer or an elderly person to detect falls.
- Different numbers of sensors may be needed to obtain the signals to compute a single parameter (e.g., at least three sensors are required to compute the electrocardiogram or EKG).
- Sensors need to be positioned in different locations on the body to acquire the necessary signals (e.g., sensors for EKG go in three different locations on the body, whereas pulse-ox sensors and accelerometers go in other locations on the body).
- Different subsets of sensors and devices may be used at different times, necessitating their easy attachment and removal, or plug and play. For instance, the gamer may want to record how his body feels and reacts while being immersed in the game and, at other times, may also want to record his experience.
- The signals from the various sensors and in different physical locations (such as first responders responding to a disaster scene) have to be sensed, collected, processed, stored, and transmitted to the remote control and coordination location.
- Signals from different types of sensors (e.g., body temperature, EKG, accelerometers) have to be processed in parallel to evaluate the various parameters in real-time.
- Since a large number of sensors is usually required, these sensors would have to be low cost and hence would likely have minimal built-in (onboard) processing capabilities.
- The sensors should be power-aware (i.e., have low power requirements).
- Power must be supplied (distributed) to the various sensors and processors.
- Thus, there is a need for a platform that has both a physical form factor and an integrated information infrastructure.

The Future of Wearables : Defining the Research Roadmap

- The paradigm of "Information Anywhere, Anytime, Anyone" is a reality today.
- For instance, a racing car enthusiast in Cupertino, California can see on his mobile device the driver's view of the track as he negotiates the Daytona Speedway.
- He can instantly access all the "stats" associated with the lap, the race, the standing, the history, and so on, thanks to the convergence of high-performance computing, communications, video, and data fusion technologies.



Need for Wearable Haptic Devices

- The field of haptics has grown in recent decades, and haptic display is now viewed as a highly promising approach to interfaces in human – computer interaction because it provides users with nuanced information in a virtual environment.
- In applications ranging from physical rehabilitation to teleoperation of remote robotic systems, haptic feedback can provide intuitive and valuable information, thereby facilitating the user's understanding of force and motion.
- The word haptic comes from the Greek word haptikos, which relates to the sense of touch – also known as the tactile sense .
- Humans rely on the sense of touch in real environments. This sense allows us to understand the nature of objects.
- The goal of a haptic feedback system is to help users feel physical characteristics, interact with virtual or distant environments, and intuitively manipulate remote objects.
- Humans utilize tactile and haptic information to determine object properties using thousands of mechanoreceptors located in the hands.
- Tactile and haptic information is essential for completing tasks and control operations.
- Haptic feedback systems can be classified into three types according to their mechanical grounding configuration:

1. Grounded Type
2. Non-grounded Type
3. Wearable Type

- **Grounded Haptic Devices** readily provide information on weight sensation and three dimensional forces.
- PHANToM is one of the most widely used commercially available haptic devices.
- It has a serial link mechanism with a counterbalance weight and a back-drivable arm, and can exert forces along multiple axes and torques at the device's pen tip.
- Sato et al. developed a series of wire-driven haptic devices (the SPIDAR series) such as SPIDAR-G, which has seven degrees of freedom (DOF) for force feedback: three for translation, three for rotation and one for grasping.
- The device provides smooth force feedback with minimal inertia, no backlash, and superior levels of scalability and safety.
- Arata et al. developed a haptic device that can exert three-axis force control (DELTA-4). Its parallel link mechanism contributes to the high rigidity of the device.
- In the robotics field, various haptic feedback methods have been developed for a teleoperationsystem .
- These devices provide powerful haptic feedback with a simplified design.
- Their weight is of little significance because they are usually fixed to a static object or to the ground.

- However, haptic interaction with virtual environments is limited to small workspaces.
- **Non-grounded Haptic Devices** employ a counteractive mechanism that creates linear or angular momentum and does not require ground connection, thus allowing for the device to be mobile or wearable.
- However, such devicescannot generate large forces. Exerting force in multiple directions and maintaining continuous intensity are also problematic.
- **Wearable Haptic Devices** allow for greater freedom and a larger workspace than grounded systems.
- As the computing environment moves from desktop to mobile platforms, new computer interfaces and human–computer interaction tools for mobile systems are required.
- Wearable sensor systems have been developed to allow monitoring of people anywhere and at any time and wearable haptics can be utilized as a part of the feedback mechanism.
- Because tactile stimulation can be perceived through the skin all over the body, utilizing the sense of touch is key in maximizing attentional resources.
- A device that can be strapped to the user is desirable for the provision of more intuitive sensation because placement near the user's skin and/or joints enables more direct haptic feedback.

Categories of Wearable Haptic

- Force Feedback
- Vibro-Tactile Feedback
- Electronic Feedback
-

Force Feedback Devices

- Force feedback provides stimuli for the kinematic sense.
- There are two types of haptic feedback systems with force feedback capability that can be considered as wearable devices: exoskeletons and fingertip-mounted devices.

Vibro-tactile feedback devices

- Vibro-tactile feedback is more aimed at the tactile senses.
- Vibration is widely used as a haptic technology for cutaneous feedback to stimulate the tactile senses.
- Haptic display based on vibrotactile feedback is easy to implement and provides robust sensation.
- Such feedback is a useful tool for guiding users with limited visual cues.

Electro-tactile feedback devices

- Electronic feedback leverages the electrical conductivity of human skin to evoke tactile sensations.
- Electro-tactile stimulation evokes tactile sensation on the skin by passing a current through surface electrodes.
- Although electro-tactile display requires direct contact between the skin and the surface electrode, small flexible electrodes are now commercially available.

Staff Incharge **HOD**